Kolmogorov-Arnold Measure Transport for Continuous Graphical Models

Stefano Fochesatto and Google Gemini

November 11, 2026

Abstract

In the Localized Sparsity Identification for Non-Gaussian Distributions (L-SING) framework [7], undirected graphical models are discovered by estimating local conditional transport maps $S^k(x_{-k}, \cdot)$ that push forward continuous conditional distributions $\pi(x_k \mid x_{-k})$ to a standard normal reference measure $\eta_k = \mathcal{N}(0, 1)$. Existing implementations parameterize $S^k(x)$ using Unconstrained Monotonic Neural Networks (UMNNs) [15], which enforce target monotonicity $\partial_k S^k > 0$ via numerical quadrature (e.g., 50-node Clenshaw–Curtis integration), imposing substantial computational overhead and requiring higher-order automatic differentiation through the discretized integral to evaluate the generalized precision matrix. In this work, we develop a certified theory and analytical implementation of Monotonic Kolmogorov–Arnold Network (KAN) conditional transport maps for L-SING. First, we formalize four admissibility criteria dictated by measure transport and generalized precision recovery: uniform target slope positivity $\partial_k S^k(x) \geq \epsilon > 0$, global $\mathbb{R}^d$ domain extrapolation, C^3 regularity for third-order mixed log-density derivatives $\partial_j \partial_k^2 S^k(x)$, and unconstrained conditioning freedom along x_{-k}. Second, we prove that strict target monotonicity is structurally enforceable across single-layer, two-layer, and arbitrarily deep (L-layer) KANs for any univariate basis family admitting a discrete-to-continuum monotonicity transfer, and we introduce Scale-Modulated Monotonic KANs (SM-KANs) that resolve the algebraic polarization obstruction of purely additive monotonic networks on multiplicative and heteroscedastic non-Gaussian couplings. Third, we instantiate this framework with monotonic B-spline bases that fulfill the higher-order regularity required for generalized precision evaluation. Because these parameterizations represent the transport map directly rather than through a discretized integral, all partial derivatives required by the generalized precision matrix admit exact closed-form evaluation, and we further show that multiplicative scale modulation in SM-KANs introduces no additional third-order curvature terms into the log-Jacobian component of the precision matrix. Empirical benchmarks across Gaussian graphical models, multiplicative Butterfly couplings, mixed non-Gaussian benchmarks, and 156-gene TCGA Ovarian Cancer networks demonstrate that monotonic KAN transport maps achieve 84× to 300× faster precision matrix extraction and 284× lower memory consumption than UMNN baselines while improving test log-likelihood and graph recovery accuracy.

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1 Introduction and Problem Setting

Probabilistic graphical models represent the conditional independence relationships among a collection of continuous random variables $X = (X_1, \ldots, X_d) \sim \pi$ via an undirected graph $G = (V, E)$, where $V = \{1, \ldots, d\}$ and edges $(j, k) \in E$ indicate direct pairwise conditional dependence. For Gaussian distributions, conditional independence is completely characterized by the sparsity pattern of the precision (inverse covariance) matrix Σ^{-1},

$$\Sigma^{-1}_{jk} = 0 \iff X_j \perp X_k \mid X_{-jk}. \tag{1}$$

However, continuous real-world systems, such as gene regulatory networks, financial markets, and geophysical flows, regularly exhibit skewness, heavy tails, multi-modality, and nonlinear heteroscedastic coupling, rendering parametric Gaussian estimators such as the Graphical Lasso (GLASSO) [3] and semi-parametric Gaussian copula transforms [8] inadequate.

To recover graphical model structure without Gaussianity assumptions, Baptista et al. [2] and Morrison et al. [11] established the Sparsity Identification in Non-Gaussian Distributions (SING) framework using measure transport. For any smooth positive density π, pairwise conditional independence is encoded by the vanishing of the mixed partial derivative of the log-density, $\partial_j \partial_k \log \pi(x) = 0$ for all $x \in \mathbb{R}^d$. SING estimates this quantity by constructing a joint lower-triangular transport map $S : \mathbb{R}^d \to \mathbb{R}^d$ pushing π forward to the standard normal density $\mathcal{N}(0, I_d)$. While theoretically general, estimating a joint triangular map requires parameterizing and coupling $\mathcal{O}(d^2)$ directional components simultaneously, which becomes computationally prohibitive as the dimension d grows.

To overcome this bottleneck, Liaw et al. [7] introduced **L-SING** (Localized Sparsity Identification for Non-Gaussian Distributions), which exploits the local Markov property,

$$\pi(x_k \mid x_{-k}) = \pi(x_k \mid x_{\mathrm{Nb}(k)}), \tag{2}$$

stating that each variable X_k is conditionally independent of all non-neighboring variables $X_{V \setminus (\{k\} \cup \mathrm{Nb}(k))}$ given its neighborhood $X_{\mathrm{Nb}(k)}$. L-SING decouples the global triangular problem into d independent, parallelizable univariate conditional transport maps $S^k : \mathbb{R}^d \to \mathbb{R}$, $x \mapsto S^k(x_{-k}, x_k)$, each pushing the conditional distribution $\pi(x_k \mid x_{-k})$ forward to a univariate standard normal reference $\eta_k = \mathcal{N}(0, 1)$.

Despite this node-wise parallelization, existing L-SING implementations rely on Unconstrained Monotonic Neural Networks (UMNNs) [15] to enforce $\partial_k S^k(x_{-k}, x_k) > 0$. Because UMNNs define S^k implicitly through numerical quadrature of a strictly positive integrand network over a Q-node Clenshaw–Curtis grid, every forward evaluation requires Q network passes, and extracting the generalized precision score $\Omega_{jk} = \mathbb{E}_\pi \left| \partial_j \partial_k \log \pi(X_k \mid X_{-k}) \right|$ requires nested higher-order automatic differentiation through the discretized integral up to third order ($\partial_j \partial_k^2 S^k$). In this manuscript, we replace quadrature-based UMNNs with analytically differentiable Monotonic Kolmogorov–Arnold Networks (KANs) [10, 12]. By equipping univariate KAN edges with basis families that transfer discrete algebraic constraints into continuous monotonicity, specifically B-splines with ordered control points, we obtain a closed-form parameterization and analytical derivatives up to third order, certified monotonicity of conditional transport maps, and orders-of-magnitude reductions in precision extraction time and memory footprint.

2 The Measure Transport Formulation in L-SING

Remark 2.1 (Context and Scope). *Note for collaborator:* For completeness and clarity of notation, the following subsections provide a self-contained re-derivation of the L-SING conditional pullback density formula and the resulting learning loss function.

2.1 Conditional Transport Maps

Let $\eta_k = \mathcal{N}(0,1)$ denote the standard univariate Gaussian reference measure with Lebesgue density,

$$\eta_k(z_k) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} z_k^2\right). \tag{3}$$

For each node $k \in \{1, \dots, d\}$, L-SING seeks a conditional transport map component $S^k(x_{-k}, \cdot) : \mathbb{R} \to \mathbb{R}$, $x_k \mapsto z_k = S^k(x_{-k}, x_k)$, that pushes forward the conditional probability measure $\nu_{\pi(\cdot|x_{-k})}$ to the standard normal reference measure ν_{η_k},

$$S^k(x_{-k}, \cdot)_\sharp \, \nu_{\pi(\cdot|x_{-k})} = \nu_{\eta_k}. \tag{4}$$

By definition of the pushforward measure in set-integral form, for any Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}\Big(Z_k \in B \;\Big|\; X_{-k} = x_{-k}\Big) = \mathbb{P}\Big(X_k \in (S^k(x_{-k}, \cdot))^{-1}(B) \;\Big|\; X_{-k} = x_{-k}\Big). \tag{5}$$

Expressed as Lebesgue integrals against their respective probability densities,

$$\int_B \eta_k(z_k)\, dz_k = \int_{(S^k(x_{-k},\cdot))^{-1}(B)} \pi(x_k \mid x_{-k})\, dx_k \quad \forall B \in \mathcal{B}(\mathbb{R}). \tag{6}$$

[Key Requirement: 1. Target Monotonicity $\partial_k S^k > 0$]

For the fiber map $x_k \mapsto S^k(x_{-k}, x_k)$ to be an orientation-preserving diffeomorphism on $\mathbb{R}$, its derivative along the target coordinate must be strictly positive everywhere,

$$\partial_k S^k(x) := \frac{\partial S^k(x)}{\partial x_k} > 0 \quad \forall x \in \mathbb{R}^d. \tag{7}$$

This ensures that the coordinate transformation $x_k \mapsto z_k$ is one-to-one, globally invertible on $\mathbb{R}$, and has a strictly positive Jacobian $|\partial_k S^k(x)| = \partial_k S^k(x) > 0$.

Under this strict monotonicity condition, we apply the 1D change-of-variables substitution $z_k = S^k(x_{-k}, x_k)$ to the right-hand side of (6). As x_k traverses $(S^k(x_{-k}, \cdot))^{-1}(B)$, z_k traverses B, with infinitesimal element $dz_k = \partial_k S^k(x_{-k}, x_k)\, dx_k$. Substituting $dx_k = \frac{1}{\partial_k S^k(x_{-k}, (S^k(x_{-k},\cdot))^{-1}(z_k))}\, dz_k$ yields,

$$\int_B \eta_k(z_k)\, dz_k = \int_B \frac{\pi\Big((S^k(x_{-k}, \cdot))^{-1}(z_k) \;\Big|\; x_{-k}\Big)}{\partial_k S^k\Big(x_{-k}, (S^k(x_{-k}, \cdot))^{-1}(z_k)\Big)}\, dz_k. \tag{8}$$

Rearranging into a single integral over B,

$$\int_B \left(\eta_k(z_k) - \frac{\pi\Big((S^k(x_{-k}, \cdot))^{-1}(z_k) \;\Big|\; x_{-k}\Big)}{\partial_k S^k\Big(x_{-k}, (S^k(x_{-k}, \cdot))^{-1}(z_k)\Big)}\right) dz_k = 0. \tag{9}$$

Since this identity must hold for **every** Borel set $B \in \mathcal{B}(\mathbb{R})$, the integrand vanishes Lebesgue-almost everywhere,

$$\eta_k(z_k) = \frac{\pi\Big((S^k(x_{-k}, \cdot))^{-1}(z_k) \;\Big|\; x_{-k}\Big)}{\partial_k S^k\Big(x_{-k}, (S^k(x_{-k}, \cdot))^{-1}(z_k)\Big)}. \tag{10}$$

Substituting $z_k = S^k(x)$ and solving for the conditional density yields the exact **conditional pullback density formula**,

$$\pi_{S^k}(x_k \mid x_{-k}) = \eta_k(S^k(x)) \cdot |\partial_k S^k(x)| = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(S^k(x))^2\right) \cdot \partial_k S^k(x). \tag{11}$$

Taking the natural logarithm yields the conditional log-density of the model,

$$\log \pi_{S^k}(x_k \mid x_{-k}) = -\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x) - \frac{1}{2}\log(2\pi). \tag{12}$$

2.2 Connection to the Learning Objective and Functional Regularization

The training objective in L-SING is derived directly from minimizing the Kullback–Leibler (KL) divergence between the true data distribution π and the model's pullback distribution π_{S^k},

$$D_{\mathrm{KL}}(\pi \,\|\, \pi_{S^k}) = \mathbb{E}_{X\sim\pi}\left[\log \pi(X_k \mid X_{-k}) - \log \pi_{S^k}(X_k \mid X_{-k})\right] \tag{13}$$

$$= -\mathbb{E}_{X\sim\pi}\left[\log \pi_{S^k}(X_k \mid X_{-k})\right] + \text{const}, \tag{14}$$

where the constant depends only on the true data entropy and does not involve the map parameters. Substituting the conditional log-pullback formula (12) into this expected negative log-likelihood directly yields,

$$-\mathbb{E}_{X\sim\pi}\left[\log \pi_{S^k}(X_k \mid X_{-k})\right] = \mathbb{E}_{X\sim\pi}\left[\frac{1}{2}(S^k(X))^2 - \log \partial_k S^k(X)\right] + \frac{1}{2}\log(2\pi). \tag{15}$$

Dropping the irrelevant additive constant $\frac{1}{2}\log(2\pi)$ and replacing the expectation with an empirical Monte Carlo average over M i.i.d. samples $\{x^{(i)}\}_{i=1}^M \sim \pi$, we obtain the empirical negative log-likelihood. To promote sparsity, a regularization term $\Phi(S^k)$ is added, leading to the regularized loss function,

$$\mathcal{L}(S^k) \coloneqq \underbrace{\frac{1}{M}\sum_{i=1}^{M}\left[\frac{1}{2}(S^k(x^{(i)}))^2 - \log \partial_k S^k(x^{(i)})\right]}_{\text{Empirical Negative Log-Pullback Likelihood } -\frac{1}{M}\sum_i \log \pi_{S^k}(x_k^{(i)} | x_{-k}^{(i)})} + \lambda\Phi(S^k), \tag{16}$$

which is minimized over the space of admissible maps, $\min_{S^k\in\mathcal{S}_k} \mathcal{L}(S^k)$, subject to the strict target monotonicity constraint $\partial_k S^k(x) > 0$ (π-a.e.). The regularization penalty $\Phi(S^k)$ is defined as,

$$\Phi(S^k) \coloneqq \sum_{j=1}^{d} \sqrt{\frac{1}{M}\sum_{i=1}^{M}\left(\frac{\partial S^k(x^{(i)})}{\partial x_j}\right)^2}. \tag{17}$$

2.3 The Generalized Precision Matrix

Spantini, Bigoni, and Marzouk [14] proved that for distributions with twice-differentiable log-densities, pairwise conditional independence is characterized by the vanishing of the mixed second derivatives of the log-density,

$$X_j \perp X_k \mid X_{-jk} \iff \partial_j\partial_k \log \pi(x) = 0 \quad \forall x \in \mathbb{R}^d. \tag{18}$$

Because $\partial_j\partial_k \log \pi(x) = \partial_j\partial_k \log \pi(x_k \mid x_{-k})$, the entries of the *generalized precision matrix* $\Omega \in \mathbb{R}^{d\times d}$ are defined as,

$$\Omega_{jk} \coloneqq \mathbb{E}\,|\partial_j\partial_k \log \pi(x_k \mid x_{-k})| = \mathbb{E}\left|\partial_j\partial_k\left[-\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x)\right]\right|. \tag{19}$$

Using N independent evaluation samples, the Monte Carlo estimator is,

$$\widehat{\Omega}_{jk} \coloneqq \frac{1}{N}\sum_{i=1}^{N}\left|\partial_j\partial_k\left[-\frac{1}{2}(S^k(x^{(i)}))^2+\log\partial_k S^k(x^{(i)})\right]\right|. \tag{20}$$

The matrix is symmetrized via $\widehat{\Omega}_{\text{L-SING}} = \frac{1}{2}(\widehat{\Omega}+\widehat{\Omega}^T)$, the diagonal is fixed to $\widehat{\Omega}_{kk}=1$, and values are normalized to a maximum of 1. Edges are determined by thresholding: $(j,k)\in E \iff |\widehat{\Omega}_{jk}|>\tau$.

[Key Requirement: 2. Higher Regularity (C^3 Smoothness)]

Evaluating the generalized precision entries $\Omega_{jk} = \mathbb{E}|\partial_j\partial_k \log\pi(x_k \mid x_{-k})|$ requires expanding the log-derivative integrand,

$$\begin{aligned}\partial_j\partial_k\left[-\frac{1}{2}(S^k(x))^2+\log\partial_k S^k(x)\right] &= -(\partial_j S^k)(\partial_k S^k) - S^k(\partial_j\partial_k S^k)\\ &\quad + \frac{(\partial_j\partial_k^2 S^k)(\partial_k S^k)-(\partial_k^2 S^k)(\partial_j\partial_k S^k)}{(\partial_k S^k)^2}.\end{aligned} \tag{21}$$

Consequently, the transport map S^k must possess continuous derivatives up to third order ($S^k\in C^3(\mathbb{R}^d)$, or piecewise C^3) so that the mixed third derivative $\partial_j\partial_k^2 S^k$ is well-defined.

[Key Requirement: 3. Efficient Derivative Computability]

Evaluating (21) across N Monte Carlo evaluation samples and all d^2 pairs (j,k) must be computationally tractable. Under numerical quadrature architectures (such as UMNNs), this requires backpropagating three times through a numerical integrator, creating massive memory footprints and runtime bottlenecks. The transport map should ideally admit closed-form, analytic derivatives to allow direct, instantaneous precision matrix extraction.

3 Limitations of Existing Architectures: MLPs and UMNNs

3.1 The Convexity Trap of Monotonic MLPs

In standard Multi-Layer Perceptrons (MLPs), enforcing monotonicity with respect to an input x_k requires constraining network weights along that input's computational path to be non-negative ($W\geq 0$) and using non-decreasing activation functions (e.g., ReLU or Sigmoids). However, a well-known result by Archer and Wang [1] proves that constraining MLP weights to be non-negative with convex activation functions forces the entire neural network to be **globally convex** [9].

This creates a fundamental bottleneck for transport maps: a convex function cannot model sigmoidal transitions, S-curves, multi-modal density transformations, or higher-order inflections without introducing severe model bias (see Figure 1).

3.2 The Quadrature Bottleneck of UMNNs

To circumvent the convexity trap, the baseline L-SING implementation parameterizes $S^k(x)$ using Unconstrained Monotonic Neural Networks (UMNNs) [15],

$$S^k(x) = \int_0^{x_k} f(t, x_{-k};\psi)\,dt + \beta(x_{-k}), \tag{22}$$

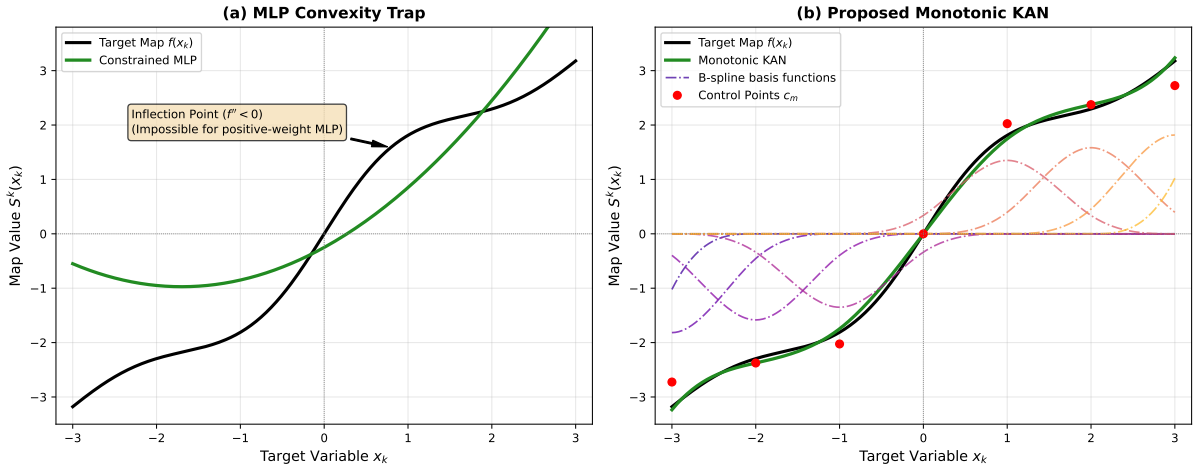


Figure 1: **Function Expressivity and Spline Parameterization. (a)** The MLP convexity trap: positive-weight MLPs [1] are constrained to convex curves ($f'' \geq 0$), failing to fit non-convex inflections of the target map $f(x_k)$. **(b)** Monotonic KAN representation: the learned spline $S^k(x_k)$ (green) fits the target map $f(x_k)$ (black), parameterized by strictly increasing control points c_m (red dots) and the superposition of B-spline basis functions.

where $f : \mathbb{R}^d \to \mathbb{R}^+$ is an unconstrained neural network strictly constrained to positive outputs via $\exp(\cdot)$ or $\text{softplus}(\cdot)$.

While UMNNs guarantee $\partial_k S^k(x) = f(x; \psi) > 0$ strictly by construction, evaluating $S^k(x)$ requires numerical integration,

$$S^k(x) \approx x_k \sum_{m=1}^{N_{\text{quad}}} w_m f\left(x_k \tau_m, x_{-k}; \psi\right), \tag{23}$$

typically using a 50-node Clenshaw–Curtis quadrature grid per sample. This introduces several major computational drawbacks:

1. **Evaluation Overhead:** Each forward evaluation of $S^k(x)$ requires $N_{\text{quad}} = 50$ forward passes through the network f, increasing training and evaluation runtimes by orders of magnitude.

2. **Nested Autodiff Through Quadrature:** Computing the Rosasco penalty $\Phi(S^k)$ requires taking first derivatives $\partial_j S^k$, requiring backpropagation through the 50-point numerical summation. Computing the generalized precision matrix entries $\widehat{\Omega}_{jk}$ requires third-order automatic differentiation through the numerical quadrature graph, leading to massive GPU/CPU memory consumption and slow extraction times (e.g., $>$ 200 seconds for $d = 40$).

4 Admissibility Requirements for the Transport Map S^k

To replace the UMNN parameterization, we must formalize the exact conditions that an admissible transport map S^k must fulfill.

Definition 4.1 (Transport Map Admissibility Criteria)**.** A parameterization of $S^k : \mathbb{R}^d \to \mathbb{R}$ is admissible for L-SING if and only if it satisfies the following criteria:

1. **Strict Target Positivity / Invertibility (Key Requirement 1):**

$$\partial_k S^k(x) \coloneqq \frac{\partial S^k(x)}{\partial x_k} \geq \epsilon > 0 \quad \forall x \in \mathbb{R}^d. \tag{24}$$

This ensures that $S^k(x_{-k}, \cdot)$ is a 1D orientation-preserving diffeomorphism on $\mathbb{R}$, guaranteeing the existence of a unique inverse pushforward and preventing numerical divergence $\log \partial_k S^k(x) \to -\infty$.

2. **Global Real-Line Domain ($\mathbb{R}^d$ Extrapolation):** The condition $\partial_k S^k(x) \geq \epsilon > 0$ must hold everywhere on $\mathbb{R}^d$, including outside the compact empirical training domain or spline knot interval $[t_{\min}, t_{\max}]$.

3. **Higher Regularity / C^3 Smoothness (Key Requirement 2):** To evaluate the generalized precision integrand,

$$T_{jk}(x) = \partial_j \partial_k \left[-\frac{1}{2}(S^k(x))^2 + \log \partial_k S^k(x) \right], \tag{25}$$

S^k must possess continuous derivatives up to order 3 ($S^k \in C^3(\mathbb{R}^d)$, or piecewise continuous 3rd derivatives), ensuring that the mixed derivative $\partial_j \partial_k^2 S^k$ is well-defined.

4. **Conditioning Freedom (Unconstrained on x_{-k}):** For all $j \neq k$, the derivative $\partial_j S^k(x)$ must be **unconstrained in sign**. The conditioning relationship must be capable of learning arbitrary, non-monotonic, multimodal responses to faithfully capture complex conditional distributions $\pi(x_k \mid x_{-k})$.

Remark 4.2 (Mandatory Requirements and Desirable Properties). Beyond these mandatory requirements, an even further nice-to-have is **efficient derivative computability** (Key Requirement 3). In Section 5, we turn to Kolmogorov–Arnold Networks (KANs), establishing a parameterization framework that not only certifies all four mandatory admissibility criteria by construction, but also natively delivers closed-form, analytic derivative evaluations.

5 Theoretical Framework and Monotonicity Proofs for KANs

The architectural foundation of Kolmogorov–Arnold Networks (KANs) [10] is rooted in the Kolmogorov–Arnold representation theorem [6], which establishes that any continuous multivariate function on a compact domain can be expressed as a finite superposition of continuous univariate functions and additions.

Theorem 5.1 (Kolmogorov–Arnold Representation Theorem [6])**.** *Let $f : [0,1]^d \to \mathbb{R}$ be an arbitrary continuous multivariate function. Then f admits the exact representation,*

$$f(x_1, \ldots, x_d) = \sum_{q=1}^{2d+1} \Phi_q \left(\sum_{p=1}^{d} \phi_{q,p}(x_p) \right), \tag{26}$$

where $\Phi_q : \mathbb{R} \to \mathbb{R}$ and $\phi_{q,p} : [0,1] \to \mathbb{R}$ are continuous univariate functions.

While the classical theorem proves representation existence with functions that may be fractal or non-smooth, modern KANs (Liu et al. [10]) operationalize this principle by parameterizing the univariate edge functions with smooth, continuously differentiable splines (such as B-splines) and generalizing the structure to arbitrary network depths $L \geq 2$ and layer widths.

As illustrated in Figure 2, KANs invert the classical MLP design: where MLPs place fixed non-linear activations σ on nodes and learnable linear weights on edges, KANs place learnable 1D spline functions ϕ directly on edges and simple summations $\sum$ at nodes. Crucially, each KAN layer $l \in \{1, \ldots, L\}$ with n_{l-1} inputs and n_l outputs forms a matrix of 1D functions,

$$\mathbf{\Phi}_l = \begin{pmatrix} \phi_{l,1,1}(\cdot) & \cdots & \phi_{l,1,n_{l-1}}(\cdot) \\ \vdots & \ddots & \vdots \\ \phi_{l,n_l,1}(\cdot) & \cdots & \phi_{l,n_l,n_{l-1}}(\cdot) \end{pmatrix}, \tag{27}$$

updating node activations via $x_j^{(l)} = \sum_{i=1}^{n_{l-1}} \phi_{l,j,i}(x_i^{(l-1)})$ from input $x^{(0)} = x$. A deep KAN of depth $L \geq 2$ is then simply the functional composition $\mathrm{KAN}(x) = (\mathbf{\Phi}_L \circ \cdots \circ \mathbf{\Phi}_1)(x)$, directly bridging the classical 2-layer theorem to deep expressive representations.

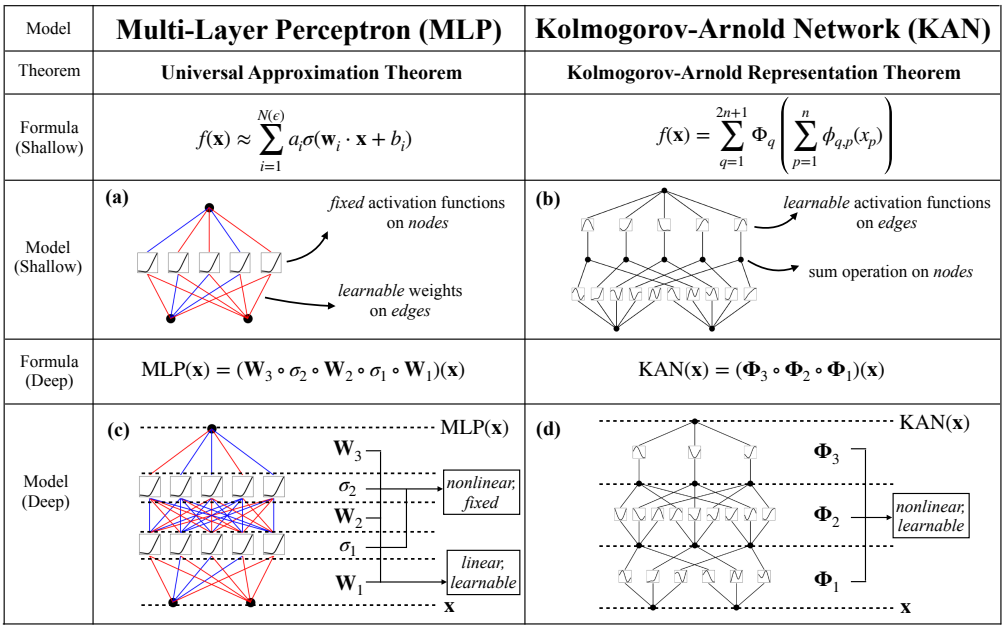


Figure 2: **Architectural Comparison: Multi-Layer Perceptrons (MLPs) vs. Kolmogorov–Arnold Networks (KANs)** (reproduced from Liu et al. [10], Figure 0.1). **(a) MLP Layer:** Fixed non-linear activation functions $\sigma(\cdot)$ (e.g., ReLU, SiLU) reside on nodes (neurons), while learnable scalar linear weights W reside on edges, computing $x^{(l)} = \sigma(W_l x^{(l-1)})$. **(b) KAN Layer:** Learnable activation functions (parameterized as smooth 1D B-splines $\phi(x)$) reside directly on edges, while receiving nodes perform simple linear addition $\sum$ without non-linearities. This 2-layer graph $[d \to 2d+1 \to 1]$ provides the direct computational realization of the classical Kolmogorov–Arnold representation theorem (26). **(c) Deep MLP Architecture:** Stacks multiple layers by interleaving affine weight matrices W and static node non-linearities: $\mathrm{MLP}(x) = (W_L \circ \sigma \circ \cdots \circ W_1)x$. **(d) Deep KAN Architecture:** Generalizes the edge-activation principle to arbitrary depths $L \geq 2$ and layer widths $[n_0, n_1, \ldots, n_L]$. Each layer transition $l \in \{1, \ldots, L\}$ applies an $n_l \times n_{l-1}$ function matrix of 1D splines $\mathbf{\Phi}_l = (\phi_{l,j,i})$ followed by summation $(x_j^{(l)} = \sum_{i=1}^{n_{l-1}} \phi_{l,j,i}(x_i^{(l-1)}))$, defining the deep functional composition $\mathrm{KAN}(x) = (\mathbf{\Phi}_L \circ \cdots \circ \mathbf{\Phi}_1)(x)$.

We now establish that the admissibility requirements of Definition 4.1 can be certified constructively for any edge basis family admitting a discrete-to-continuum monotonicity transfer. We first state this foundational property, demonstrate why purely additive ($L = 1$) networks are structurally insufficient for conditional independence discovery, and then analyze the canonical two-layer ($L = 2$) KAN before generalizing to arbitrary network depths and multiplicative scale modulation.

5.1 Foundational Property: Discrete-to-Continuum Monotonicity of Edge Parameterizations

The structural monotonicity proofs for two-layer, deep, and scale-modulated KANs in this section do not depend on a specific basis expansion; rather, they rely only on the property that each univariate edge function $\phi : \mathbb{R} \to \mathbb{R}$ belongs to a parameterized family in which finite-dimensional parameter constraints certify pointwise derivative bounds across all of $\mathbb{R}$, while

admitting closed-form evaluation of ϕ and its first three derivatives. We state this requirement as a general property that will be instantiated constructively in Section 6.

Definition 5.2 (Admissible Edge Family with Discrete-to-Continuum Monotonicity)**.** A parameterized family of univariate functions $\mathcal{F} = \{\phi_\theta : \mathbb{R} \to \mathbb{R}\}_{\theta \in \Theta}$ with finite-dimensional parameter space $\Theta \subset \mathbb{R}^M$ satisfies the *discrete-to-continuum monotonicity property* if every $\phi_\theta \in \mathcal{F}$ admits closed-form evaluation of $\phi_\theta(t), \phi'_\theta(t), \phi''_\theta(t)$, and $\phi'''_\theta(t)$, and there exist parameter subsets $\Theta_{\geq 0} \subset \Theta$ and $\Theta_{\geq \epsilon} \subset \Theta$ (for any prescribed floor $\epsilon > 0$) defined by algebraic constraints on the discrete parameters $\theta \in \mathbb{R}^M$ such that:

1. **Certified Non-Decreasing Edges:** Whenever $\theta \in \Theta_{\geq 0}$, the continuous derivative satisfies
$$\phi'_\theta(t) \geq 0 \quad \forall t \in \mathbb{R}. \tag{28}$$

2. **Certified Strictly Increasing Edges:** Whenever $\theta \in \Theta_{\geq \epsilon}$, the continuous derivative satisfies the uniform positive lower bound,
$$\phi'_\theta(t) \geq \epsilon > 0 \quad \forall t \in \mathbb{R}. \tag{29}$$

3. **Unconstrained Conditioning Edges:** Whenever $\theta \in \Theta$ is left unconstrained, $\phi_\theta(t)$ and its derivatives are unrestricted in sign, slope, and curvature.

Remark 5.3 (Candidate Basis Families Satisfying Definition 5.2). Definition 5.2 encompasses a broad class of classical and spectral function parameterizations on a compact active interval $[a, b]$ extended to $\mathbb{R}$ via C^1 linear tails. Examples include degree-p B-splines and Bernstein polynomials with non-decreasing control points ($c_{m+1} - c_m \geq \delta > 0$), piecewise cubic Hermite splines subject to Fritsch–Carlson derivative constraints [5, 12], positively weighted primitives of radial basis functions (Integrated RBFs), and orthogonal polynomial expansions whose derivatives are parameterized as sum-of-squares quadratic forms. For clarity and continuity with the standard KAN literature [10], we instantiate Definition 5.2 using degree-p B-splines in Section 6.

Remark 5.4 (Why Single-Layer Additive Networks ($L = 1$) Cannot Discover General Graph Structure). Consider a purely additive single-layer transport map $S^k(x) = \phi_k(x_k) + \sum_{j \neq k} \phi_j(x_j)$. Applying Definition 5.2 to ϕ_k immediately guarantees $\partial_k S^k(x) = \phi'_k(x_k) \geq \epsilon > 0$.

However, because the conditioning coordinates enter purely additively, the target derivative $\partial_k S^k(x) = \phi'_k(x_k)$ is completely independent of x_{-k}, so all mixed cross-derivatives of the map vanish identically,

$$\partial_j \partial_k S^k(x) = \frac{\partial}{\partial x_j}[\phi'_k(x_k)] \equiv 0 \quad \text{and} \quad \partial_j \partial_k \log \partial_k S^k(x) \equiv 0 \quad \forall j \neq k. \tag{30}$$

In the generalized precision expansion (21), both the interaction term $-S^k(\partial_j \partial_k S^k)$ and the log-Jacobian coupling term vanish, reducing the integrand to the separable rank-one product $-(\partial_j S^k)(\partial_k S^k) = -\phi'_j(x_j)\phi'_k(x_k)$. Geometrically, x_{-k} can only induce a rigid location shift without modulating conditional scale, variance, or higher-order shape; for any zero-mean scale or shape coupling, $\phi_j \equiv 0$ and Ω_{jk} in (19) collapses to zero. Functional composition ($L \geq 2$) is therefore mathematically necessary to achieve $\partial_j \partial_k S^k(x) \not\equiv 0$.

5.2 Two-Layer Monotonic KAN

Having established that single-layer additive networks fail the graphical model recovery requirement, we focus on the canonical two-layer KAN $[d \to Q \to 1]$ illustrated in Figure 3. This architecture directly operationalizes the classical Kolmogorov–Arnold representation theorem (Theorem 5.1) and serves as the primary parameterization for local transport maps S^k.

Definition 5.5 (Two-Layer Monotonic KAN)**.** A 2-layer KAN transport map $S^k : \mathbb{R}^d \to \mathbb{R}$ ($L = 2$) with $n_1 = Q$ hidden summation nodes and input $x^{(0)} = x$ is parameterized via,

$$S^k(x) = x_1^{(2)} = \sum_{q=1}^{Q} \phi_{2,1,q}\big(x_q^{(1)}(x)\big), \tag{31}$$

where $x_q^{(1)}(x)$ denotes the q-th hidden activation at layer $l = 1$,

$$x_q^{(1)}(x) \coloneqq \phi_{1,q,k}(x_k) + \sum_{j \neq k} \phi_{1,q,j}(x_j), \tag{32}$$

and the edge activations satisfy:

1. **Target Inner Edges ($l = 1$):** $\phi_{1,q,k} \in \mathcal{F}$ are non-decreasing univariate edge functions ($\phi'_{1,q,k} \geq 0$), parameterized with $\theta \in \Theta_{\geq 0}$ via Definition 5.2.
2. **Outer Edges ($l = 2$):** $\phi_{2,1,q} \in \mathcal{F}$ are non-decreasing univariate edge functions ($\phi'_{2,1,q} \geq 0$, $\theta \in \Theta_{\geq 0}$), with at least one active path $\mathcal{P}^* = (k, q^*, 1)$ ($q^* \in \{1, \ldots, Q\}$) satisfying $\phi'_{1,q^*,k} \geq \epsilon_1 > 0$ ($\theta \in \Theta_{\geq \epsilon_1}$) and $\phi'_{2,1,q^*} \geq \epsilon_2 > 0$ ($\theta \in \Theta_{\geq \epsilon_2}$) via Definition 5.2.
3. **Conditioning Inner Edges ($l = 1$):** $\phi_{1,q,j} \in \mathcal{F}$ ($j \neq k$) are arbitrary unconstrained univariate edge functions ($\theta \in \Theta$).

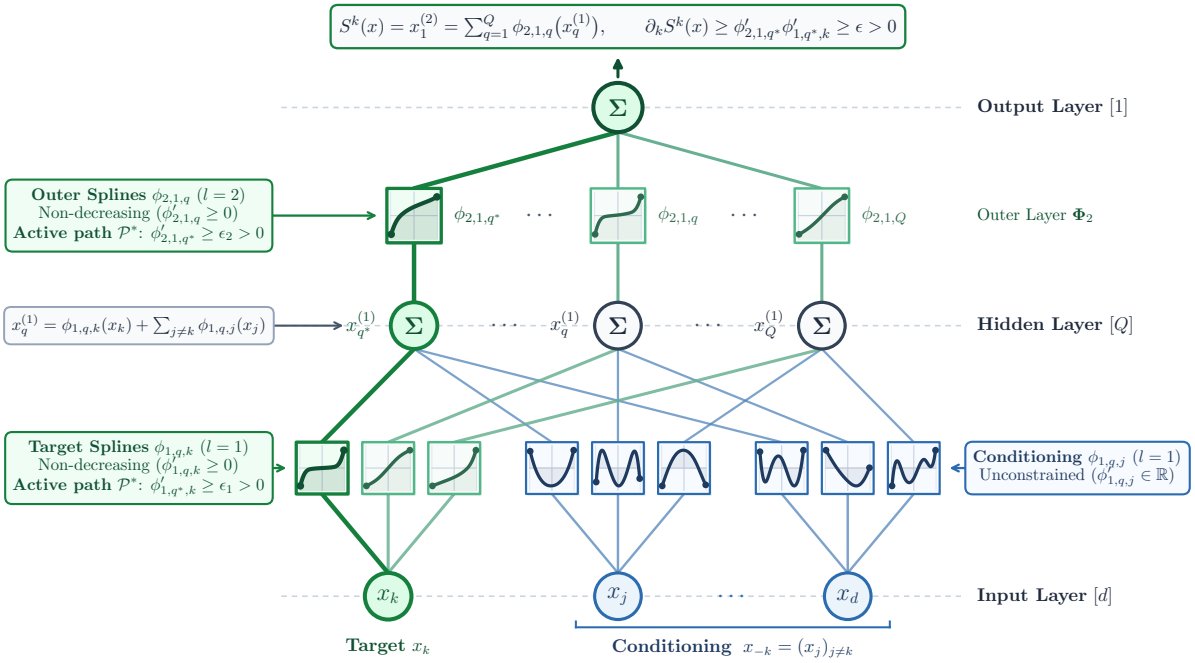


Figure 3: **Two-Layer Monotonic KAN Transport Map $S^k : \mathbb{R}^d \to \mathbb{R}$ with Topology** $[d \to Q \to 1]$ (following the edge-activation diagram convention of Liu et al. [10]). Input coordinates $x^{(0)} = x$ partition into the target feature x_k (green) and conditioning features $x_{-k} = (x_j)_{j \neq k}$ (blue), while hidden nodes $x_q^{(1)}$ and the output node $x_1^{(2)} = S^k(x)$ perform linear summations ($\sum$). Each edge carries a learnable 1D activation plot: target inner edges $\phi_{1,q,k}(x_k)$ and outer edges $\phi_{2,1,q}(x_q^{(1)})$ are non-decreasing ($\phi'_{1,q,k} \geq 0$, $\phi'_{2,1,q} \geq 0$, muted green), and the certified active path $\mathcal{P}^* = (k, q^*, 1)$ through hidden node $x_{q^*}^{(1)}$ (highlighted in bold dark green) maintains strictly positive derivatives $\phi'_{1,q^*,k} \geq \epsilon_1 > 0$ and $\phi'_{2,1,q^*} \geq \epsilon_2 > 0$, guaranteeing $\partial_k S^k(x) \geq \epsilon_1 \epsilon_2 > 0$ globally on $\mathbb{R}^d$, whereas conditioning inner edges $\phi_{1,q,j}(x_j)$ remain unconstrained (blue).

Theorem 5.6 (Admissibility and Structural Coupling in 2-Layer KANs)**.** *Let S^k be parameterized according to Definition 5.5. Then,*

1. ***Target Monotonicity (Criterion 1):*** *For all $x \in \mathbb{R}^d$, the target derivative satisfies,*

$$\partial_k S^k(x) = \sum_{q=1}^{Q} \phi'_{2,1,q}\big(x_q^{(1)}(x)\big) \cdot \phi'_{1,q,k}(x_k) \geq \epsilon > 0, \tag{33}$$

   where $\epsilon \coloneqq \epsilon_1 \epsilon_2 > 0$ is a uniform positive constant, certifying that $S^k(x_{-k}, \cdot)$ is a 1D orientation-preserving diffeomorphism on $\mathbb{R}$.

2. ***Conditioning Freedom (Criterion 4):*** *For every conditioning coordinate $j \neq k$, the conditioning partial derivatives,*

$$\partial_j S^k(x) = \sum_{q=1}^{Q} \phi'_{2,1,q}\big(x_q^{(1)}(x)\big) \cdot \phi'_{1,q,j}(x_j), \qquad \partial_j \partial_k S^k(x) = \sum_{q=1}^{Q} \phi''_{2,1,q}\big(x_q^{(1)}(x)\big) \cdot \phi'_{1,q,j}(x_j) \cdot \phi'_{1,q,k}(x_k), \tag{34}$$

   are unconstrained in sign.

Proof. Differentiating $S^k(x)$ in (31) with respect to x_k via the multivariable chain rule yields,

$$\partial_k S^k(x) = \sum_{q=1}^{Q} \frac{\partial \phi_{2,1,q}(x_q^{(1)})}{\partial x_q^{(1)}} \cdot \frac{\partial x_q^{(1)}(x)}{\partial x_k} = \sum_{q=1}^{Q} \phi'_{2,1,q}\big(x_q^{(1)}(x)\big) \cdot \phi'_{1,q,k}(x_k). \tag{35}$$

By Definition 5.2, all target inner and outer edge functions satisfy $\phi'_{1,q,k}(t) \geq 0$ and $\phi'_{2,1,q}(t) \geq 0$ globally on $\mathbb{R}$, while along the designated active path $\mathcal{P}^* = (k, q^*, 1)$ they satisfy $\phi'_{1,q^*,k}(t) \geq \epsilon_1 > 0$ and $\phi'_{2,1,q^*}(t) \geq \epsilon_2 > 0$. Decomposing the summation into the active path $\mathcal{P}^*$ and the remaining paths $q \neq q^*$ and substituting these lower bounds gives,

$$\partial_k S^k(x) = \underbrace{\phi'_{2,1,q^*}\big(x_{q^*}^{(1)}(x)\big) \cdot \phi'_{1,q^*,k}(x_k)}_{\text{Active Path } \mathcal{P}^*} + \underbrace{\sum_{q \neq q^*} \phi'_{2,1,q}\big(x_q^{(1)}(x)\big) \cdot \phi'_{1,q,k}(x_k)}_{\text{Remaining Paths}} \geq \epsilon_2 \epsilon_1 + 0 = \epsilon > 0, \tag{36}$$

for all $x \in \mathbb{R}^d$.

Next, differentiating $S^k(x)$ and $\partial_k S^k(x)$ with respect to a conditioning variable x_j $(j \neq k)$ via the chain rule yields $\partial_j S^k(x) = \sum_{q=1}^{Q} \phi'_{2,1,q}(x_q^{(1)}(x)) \cdot \phi'_{1,q,j}(x_j)$ and,

$$\begin{aligned}
\partial_j \partial_k S^k(x) &= \sum_{q=1}^{Q} \frac{\partial}{\partial x_j}\Big[\phi'_{2,1,q}\big(x_q^{(1)}(x)\big)\Big] \cdot \phi'_{1,q,k}(x_k) \\
&= \sum_{q=1}^{Q} \frac{\partial \phi'_{2,1,q}(x_q^{(1)})}{\partial x_q^{(1)}} \cdot \frac{\partial x_q^{(1)}(x)}{\partial x_j} \cdot \phi'_{1,q,k}(x_k) \\
&= \sum_{q=1}^{Q} \underbrace{\phi''_{2,1,q}\big(x_q^{(1)}(x)\big)}_{\substack{\text{Curvature of shared} \\ \text{outgoing edge}}} \cdot \underbrace{\phi'_{1,q,j}(x_j)}_{\substack{\text{Slope of incoming} \\ \text{conditioning edge}}} \cdot \underbrace{\phi'_{1,q,k}(x_k)}_{\substack{\text{Slope of incoming} \\ \text{target edge}}},
\end{aligned} \tag{37}$$

establishing (34). Because $\phi_{1,q,j}$ is unconstrained ($\theta \in \Theta$ in Definition 5.2), $\phi'_{1,q,j}(x_j)$ and $\phi''_{2,1,q}(x_q^{(1)}(x))$ can take arbitrary real values of either sign while maintaining $\phi'_{2,1,q} \geq 0$, so $\partial_j S^k(x)$ and $\partial_j \partial_k S^k(x)$ are unconstrained in sign, establishing Criterion 4. □

5.3 Arbitrarily Deep Multi-Layer KAN (L Layers)

We now generalize the construction to a KAN of arbitrary depth $L \geq 2$, with layer dimensions $[n_0, n_1, n_2, \ldots, n_L]$, where $n_0 = d$ (input features) and $n_L = 1$ (scalar output $S^k(x)$), as illustrated in Figure 4.

Definition 5.7 (Arbitrarily Deep Monotonic KAN)**.** An L-layer KAN transport map $S^k : \mathbb{R}^d \to \mathbb{R}$ ($L \geq 2$) with layer widths $[n_0, n_1, \ldots, n_L]$ ($n_0 = d$, $n_L = 1$) and input $x^{(0)} = x$ is parameterized via,

$$S^k(x) = x_1^{(L)} = \Big(\mathbf{\Phi}_L \circ \mathbf{\Phi}_{L-1} \circ \cdots \circ \mathbf{\Phi}_1\Big)(x), \tag{38}$$

where $x_j^{(l)}(x)$ denotes the j-th activation at layer $l \in \{1, \ldots, L\}$,

$$x_j^{(l)}(x) \coloneqq \sum_{i=1}^{n_{l-1}} \phi_{l,j,i}\big(x_i^{(l-1)}(x)\big), \quad j = 1, \ldots, n_l. \tag{39}$$

Let $\mathbb{P}_k$ denote the set of all directed computational paths $\mathcal{P} = (i_0, i_1, \ldots, i_L)$ from target input $i_0 = k$ to output $i_L = 1$, where $i_l \in \{1, \ldots, n_l\}$. The edge activations satisfy:

1. **Target Inner Edges ($l = 1$):** $\phi_{1,j,k} \in \mathcal{F}$ ($j \in \{1, \ldots, n_1\}$) are non-decreasing univariate edge functions ($\phi'_{1,j,k} \geq 0$), parameterized with $\theta \in \Theta_{\geq 0}$ via Definition 5.2.

2. **Downstream Hidden and Output Edges ($2 \leq l \leq L$):** $\phi_{l,j,i} \in \mathcal{F}$ are non-decreasing univariate edge functions ($\phi'_{l,j,i} \geq 0$, $\theta \in \Theta_{\geq 0}$), with at least one active path $\mathcal{P}^* = (i_0^*, i_1^*, \ldots, i_L^*) \in \mathbb{P}_k$ ($i_0^* = k$, $i_L^* = 1$) satisfying $\phi'_{l,i_l^*,i_{l-1}^*} \geq \epsilon_l > 0$ ($\theta \in \Theta_{\geq \epsilon_l}$) for all $l \in \{1, \ldots, L\}$ via Definition 5.2.

3. **Conditioning Inner Edges ($l = 1$):** $\phi_{1,j,i} \in \mathcal{F}$ ($i \neq k$) are arbitrary unconstrained univariate edge functions ($\theta \in \Theta$).

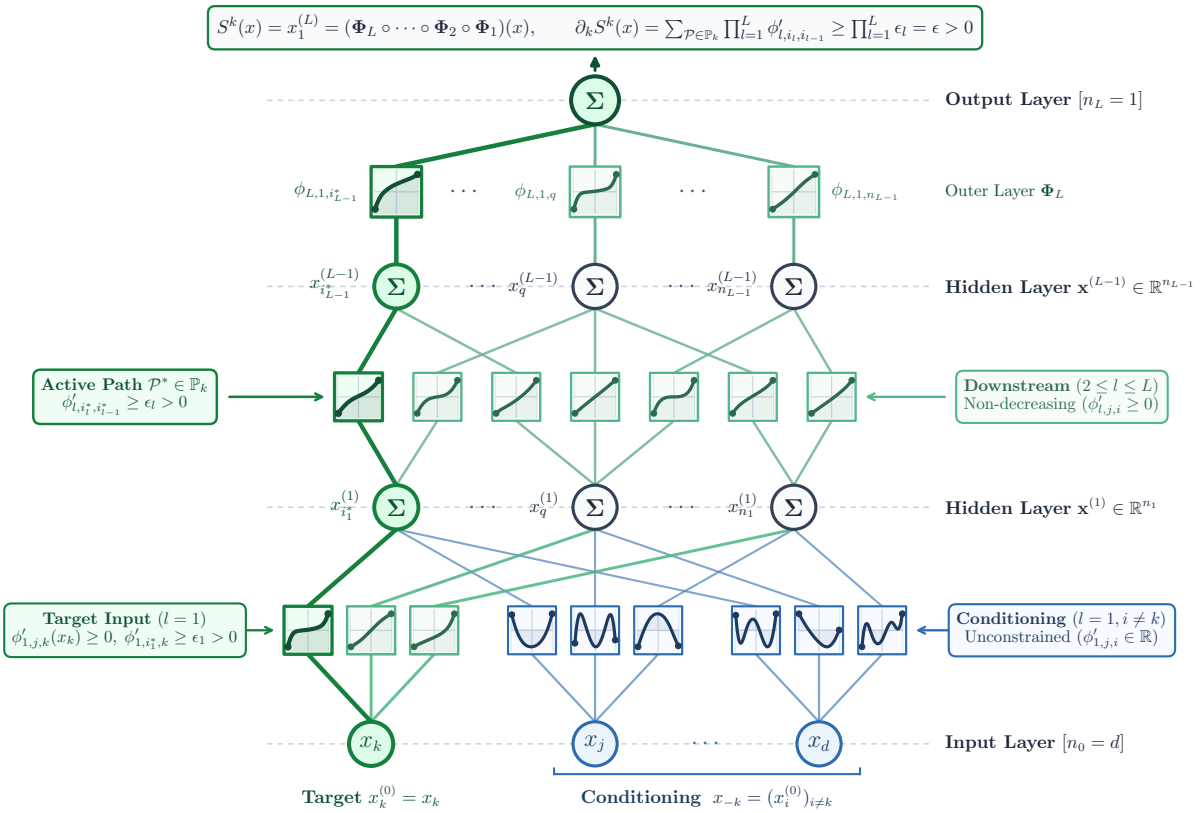


Figure 4: **Arbitrarily Deep L-Layer Monotonic KAN Transport Map** $[n_0 \to n_1 \to \cdots \to n_{L-1} \to n_L]$ $(n_0 = d, n_L = 1$; following Liu et al. [10], Figure 0.1d). Target edges emanating from x_k at layer $l = 1$ and all downstream hidden edges $(2 \leq l \leq L)$ are non-decreasing $(\phi'_{l,j,i} \geq 0$, muted green), while conditioning edges $\phi_{1,j,i}(x_i)$ $(i \neq k)$ are unconstrained (blue). At least one certified active backbone path $\mathcal{P}^* = (i_0^*, i_1^*, \ldots, i_{L-1}^*, i_L^*) \in \mathbb{P}_k$ with $i_0^* = k, i_L^* = 1$ (highlighted in bold dark green) maintains strictly positive derivatives $\phi'_{l,i_l^*,i_{l-1}^*} \geq \epsilon_l > 0$, guaranteeing $\partial_k S^k(x) \geq \prod_{l=1}^{L} \epsilon_l = \epsilon > 0$ globally on $\mathbb{R}^d$.

Theorem 5.8 (Admissibility and Certified Monotonicity for Arbitrarily Deep KANs)**.** *Let S^k be parameterized according to Definition 5.7. Then,*

1. ***Target Monotonicity (Criterion 1):*** *For all $x \in \mathbb{R}^d$, the target derivative satisfies,*

$$\partial_k S^k(x) = \sum_{\mathcal{P}\in\mathbb{P}_k} \prod_{l=1}^{L} \phi'_{l,i_l,i_{l-1}}\big(x_{i_{l-1}}^{(l-1)}\big) \geq \epsilon > 0, \tag{40}$$

   where $\epsilon := \prod_{l=1}^{L} \epsilon_l > 0$ is a uniform positive constant, certifying that $S^k(x_{-k}, \cdot)$ is a 1D orientation-preserving diffeomorphism on $\mathbb{R}$.

2. ***Conditioning Freedom (Criterion 4):*** *For every conditioning coordinate $j \neq k$, the conditioning partial derivatives $\partial_j S^k(x)$ and $\partial_j \partial_k S^k(x)$ are unconstrained in sign.*

Proof. We evaluate the sensitivity of the transport map output $S^k(x) = x_1^{(L)}$ with respect to the target input $x_k = x_k^{(0)}$. Applying the multivariable chain rule layer by layer from the output layer L back to the input layer 0 expands the total derivative into a multi-index sum,

$$\frac{\partial S^k(x)}{\partial x_k} = \frac{\partial x_1^{(L)}}{\partial x_k^{(0)}} = \sum_{i_{L-1}=1}^{n_{L-1}} \sum_{i_{L-2}=1}^{n_{L-2}} \cdots \sum_{i_1=1}^{n_1} \left(\frac{\partial x_1^{(L)}}{\partial x_{i_{L-1}}^{(L-1)}} \frac{\partial x_{i_{L-1}}^{(L-1)}}{\partial x_{i_{L-2}}^{(L-2)}} \cdots \frac{\partial x_{i_1}^{(1)}}{\partial x_k^{(0)}} \right). \tag{41}$$

From the layer definition (39), intermediate neuron activations are linear superpositions of univariate edge functions, $x_j^{(l)} = \sum_{i=1}^{n_{l-1}} \phi_{l,j,i}(x_i^{(l-1)})$. Differentiating $x_j^{(l)}$ with respect to upstream activation $x_i^{(l-1)}$, all other terms vanish, yielding the univariate edge derivative,

$$\frac{\partial x_j^{(l)}}{\partial x_i^{(l-1)}} = \phi'_{l,j,i}\big(x_i^{(l-1)}\big). \tag{42}$$

At the first layer $l = 1$, holding all conditioning coordinates $x_{-k} = \{x_i : i \neq k\}$ fixed gives $\partial x_{i_1}^{(1)} / \partial x_k^{(0)} = \phi'_{1,i_1,k}(x_k^{(0)})$.

Substituting these edge derivatives into (41), each term in the summation corresponds to the product of edge derivatives along a directed computational path $\mathcal{P} = (i_0, i_1, \ldots, i_L) \in \mathbb{P}_k$ with $i_0 = k$ and $i_L = 1$,

$$\partial_k S^k(x) = \sum_{\mathcal{P} \in \mathbb{P}_k} \prod_{l=1}^{L} \phi'_{l,i_l,i_{l-1}}\big(x_{i_{l-1}}^{(l-1)}\big). \tag{43}$$

By Definition 5.2 and Conditions 1–2 of Definition 5.7, all target-transmitting edges satisfy $\phi'_{l,j,i}(t) \geq 0$ globally on $\mathbb{R}$, so every path product in (43) is non-negative regardless of intermediate activations $x^{(l-1)}$. Isolating the certified active backbone path $\mathcal{P}^* = (i_0^*, i_1^*, \ldots, i_L^*)$, along which $\phi'_{l,i_l^*,i_{l-1}^*}(t) \geq \epsilon_l > 0$, we decompose the sum as,

$$\partial_k S^k(x) = \prod_{l=1}^{L} \phi'_{l,i_l^*,i_{l-1}^*}\big(x_{i_{l-1}^*}^{(l-1)}\big) + \sum_{\mathcal{P} \neq \mathcal{P}^*} \prod_{l=1}^{L} \phi'_{l,i_l,i_{l-1}}\big(x_{i_{l-1}}^{(l-1)}\big) \geq \prod_{l=1}^{L} \epsilon_l + 0 = \epsilon > 0. \tag{44}$$

Because this strictly positive bound holds independently of intermediate activations $x^{(l-1)}$ and does not depend on x, $S^k(x)$ is certified strictly monotonic across all of $\mathbb{R}^d$, establishing Criterion 1.

Finally, to establish Criterion 4, observe that each second-layer activation $x_{i_2}^{(2)}(x) = \sum_{i_1=1}^{n_1} \phi_{2,i_2,i_1}(x_{i_1}^{(1)}(x))$ has the exact form of a two-layer monotonic KAN (Definition 5.5). By Theorem 5.6 and (34), its conditioning partial derivatives $\partial_j x_{i_2}^{(2)}(x)$ and $\partial_j \partial_k x_{i_2}^{(2)}(x)$ with respect to any $j \neq k$,

$$\partial_j x_{i_2}^{(2)}(x) = \sum_{i_1=1}^{n_1} \phi'_{2,i_2,i_1}\big(x_{i_1}^{(1)}(x)\big) \cdot \phi'_{1,i_1,j}(x_j), \qquad \partial_j \partial_k x_{i_2}^{(2)}(x) = \sum_{i_1=1}^{n_1} \phi''_{2,i_2,i_1}\big(x_{i_1}^{(1)}(x)\big) \cdot \phi'_{1,i_1,j}(x_j) \cdot \phi'_{1,i_1,k}(x_k), \tag{45}$$

are unconstrained in sign because the first-layer conditioning edges $\phi_{1,i_1,j}$ (Condition 3 of Definition 5.7) and the curvatures ϕ''_{2,i_2,i_1} are unconstrained. Because the downstream subnetwork $(\mathbf{\Phi}_L \circ \cdots \circ \mathbf{\Phi}_3)$ transmits $x_{i_2^*}^{(2)}$ to the output $S^k(x)$ with strictly positive sensitivity $\partial S^k / \partial x_{i_2^*}^{(2)} \geq \prod_{l=3}^{L} \epsilon_l > 0$ along the active path $\mathcal{P}^*$, both $\partial_j S^k(x)$ and $\partial_j \partial_k S^k(x)$ remain unconstrained in sign, establishing Criterion 4. □

5.4 Scale-Modulated Monotonic KAN (SM-KAN)

5.4.1 The Additive Polarization Dilemma

In non-Gaussian graphical models, conditional dependencies are frequently *multiplicative* or *scale-dependent* (heteroscedastic). Consider, for example, the benchmark Butterfly distribution,

$$X_1 \sim \mathcal{N}(0,1), \quad W \sim \mathcal{N}(0,1), \quad X_1 \perp W, \quad X_2 = W \cdot X_1. \tag{46}$$

Conditioned on $X_1 = x_1 \neq 0$, the target variable follows $X_2 \mid (X_1 = x_1) \sim \mathcal{N}(0, x_1^2)$, so the exact Knothe–Rosenblatt transport map normalizing X_2 to a standard Gaussian is,

$$S^2(x_1, x_2) = \frac{x_2}{|x_1|}, \quad x_1 \neq 0. \tag{47}$$

In a purely additive 2-layer KAN (31), each hidden node combines the target edge $u_q = \phi_{1,q,k}(x_k)$ and the conditioning sum $v_q = \sum_{j \neq k} \phi_{1,q,j}(x_j)$ solely by addition, $x_q^{(1)}(x) = u_q + v_q$. Representing a multiplicative interaction $u \cdot v$ through univariate outer functions applied to sums $u \pm v$ requires algebraic polarization,

$$u \cdot v = \frac{1}{4}(u+v)^2 - \frac{1}{4}(u-v)^2. \tag{48}$$

In a standard unconstrained 2-layer KAN, (48) is realized by two branches $x_1^{(1)} = u + v$ and $x_2^{(1)} = u - v$ with quadratic outer edges $\phi_{2,1,1}(t) = \frac{1}{4}t^2$ and $\phi_{2,1,2}(t) = -\frac{1}{4}t^2$, whose first derivatives are $\phi'_{2,1,1}(t) = \frac{1}{2}t$ and $\phi'_{2,1,2}(t) = -\frac{1}{2}t$. Both outer edges therefore require negative first derivatives ($\phi'_{2,1,1}(t) < 0$ for $t < 0$ and $\phi'_{2,1,2}(t) < 0$ for $t > 0$). Under Definition 5.5, however, global target monotonicity requires $\phi'_{2,1,q}(t) \geq 0$ for all $t \in \mathbb{R}$ across every outer edge. Consequently, purely additive monotonic KANs cannot represent multiplicative conditioning without violating the non-negative outer-slope constraints required for target monotonicity.

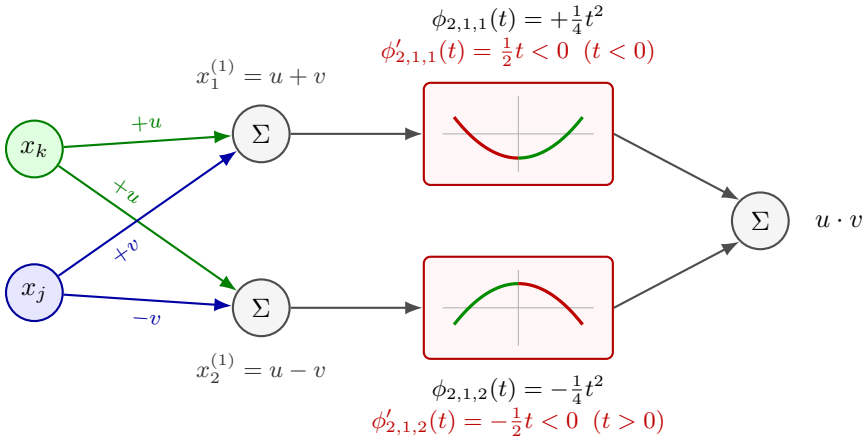


Figure 5: Realizing the polarization identity $u \cdot v = \frac{1}{4}(u+v)^2 - \frac{1}{4}(u-v)^2$ in a 2-layer KAN requires quadratic outer edges with negative slope regions (red), violating $\phi'_{2,1,q}(t) \geq 0$.

5.4.2 Scale-Modulation Architecture

To resolve this limitation without relying on polarization, we wrap the certified monotonic core KAN $S^k_{\text{core}}(x)$ in an explicit location–scale conditioning structure,

$$S^k(x) = E(x_{-k}) \cdot S^k_{\text{core}}(x) + \alpha(x_{-k}). \tag{49}$$

Here, the conditioning scale factor $E(x_{-k})$ and location shift $\alpha(x_{-k}) = \sum_{j \neq k} \rho_j(x_j)$ depend exclusively on the conditioning coordinates x_{-k}. In particular, the positive scale factor $E(x_{-k}) > 0$ captures multiplicative interactions between x_k and x_{-k}, such as conditional variance scaling or heteroscedastic noise, while strictly preserving target monotonicity.

To ensure that the scale multiplier remains strictly positive and uniformly bounded away from zero and infinity across all of $\mathbb{R}^{d-1}$ the conditioning coordinates x_{-k} are mapped to $E(x_{-k})$ through the following function,

$$E(x_{-k}) \coloneqq \exp\left(s_{\max} \tanh\left(\frac{\sigma(x_{-k})}{s_{\max}}\right)\right) \in \left(e^{-s_{\max}}, e^{s_{\max}}\right), \tag{50}$$

where $\sigma(x_{-k}) = \sum_{j \neq k} \psi_j(x_j)$ is an unconstrained additive conditioning network, and $s_{\max} > 0$ is a fixed scale ceiling (in our implementation, $s_{\max} = 4.0$). Near $\sigma(x_{-k}) \approx 0$, $s_{\max} \tanh(\sigma(x_{-k})/s_{\max}) \approx \sigma(x_{-k})$, so (50) reproduces the standard exponential scale factor $\exp(\sigma(x_{-k}))$, while globally guaranteeing $e^{-s_{\max}} < E(x_{-k}) < e^{s_{\max}}$ and preserving C^∞ smoothness everywhere on $\mathbb{R}^{d-1}$.

Placing the additive offset $\alpha(x_{-k})$ outside the product (directly analogous to the conditioning intercept $+\beta(x_{-k})$ in UMNNs (22)) allows location shifts in the $\mathcal{N}(0,1)$ reference space to be learned independently of the scale modulation. Consider, for example, a general heteroscedastic Butterfly distribution with conditional mean shift $\mu(X_1)$, where $X_2 \mid (X_1 = x_1) \sim \mathcal{N}(\mu(x_1), x_1^2)$. Writing the exact transport map as,

$$S^2(x_1, x_2) = \frac{x_2 - \mu(x_1)}{|x_1|} = \exp\big(-\log|x_1|\big) \cdot x_2 - \frac{\mu(x_1)}{|x_1|}, \quad x_1 \neq 0, \tag{51}$$

demonstrates the role of each component: the certified monotonic core learns the identity $S^2_{\text{core}}(x_1, x_2) = x_2$ ($\partial_2 S^2_{\text{core}} = 1 > 0$), while the scale network $\sigma(x_1) = \psi_1(x_1)$ approximates $-\log|x_1|$ and the offset network $\alpha(x_1) = \rho_1(x_1)$ captures the location shift $-\mu(x_1)/|x_1|$ (recovering the centered Butterfly map (47) when $\mu(x_1) \equiv 0$).

Definition 5.9 (Scale-Modulated Monotonic KAN)**.** A **Scale-Modulated Monotonic KAN (SM-KAN)** parameterizes the k-th conditional transport map $S^k : \mathbb{R}^d \to \mathbb{R}$ as,

$$S^k(x) = E(x_{-k}) \cdot S^k_{\text{core}}(x) + \alpha(x_{-k}), \tag{52}$$

where $E(x_{-k}) \coloneqq \exp\left(s_{\max} \tanh\left(\frac{\sigma(x_{-k})}{s_{\max}}\right)\right)$ with fixed scale ceiling $s_{\max} > 0$, and:

1. **Certified Monotonic Core (S^k_{core}):** $S^k_{\text{core}} : \mathbb{R}^d \to \mathbb{R}$ is an L-layer certified monotonic KAN ($L \geq 2$, parameterized according to Definition 5.5 or Definition 5.7) satisfying $\partial_k S^k_{\text{core}}(x) \geq \epsilon > 0$ for all $x \in \mathbb{R}^d$.

2. **Conditioning Scale Network (σ):** $\sigma : \mathbb{R}^{d-1} \to \mathbb{R}$ is an unconstrained additive network over the conditioning coordinates x_{-k},

$$\sigma(x_{-k}) = \sum_{j \neq k} \psi_j(x_j), \tag{53}$$

   where each $\psi_j \in \mathcal{F}$ ($j \neq k$) is an unconstrained univariate edge function.

3. **Conditioning Offset Network (α):** $\alpha : \mathbb{R}^{d-1} \to \mathbb{R}$ is an unconstrained additive network over the conditioning coordinates x_{-k},

$$\alpha(x_{-k}) = \sum_{j \neq k} \rho_j(x_j), \tag{54}$$

   where each $\rho_j \in \mathcal{F}$ ($j \neq k$) is an unconstrained univariate edge function.

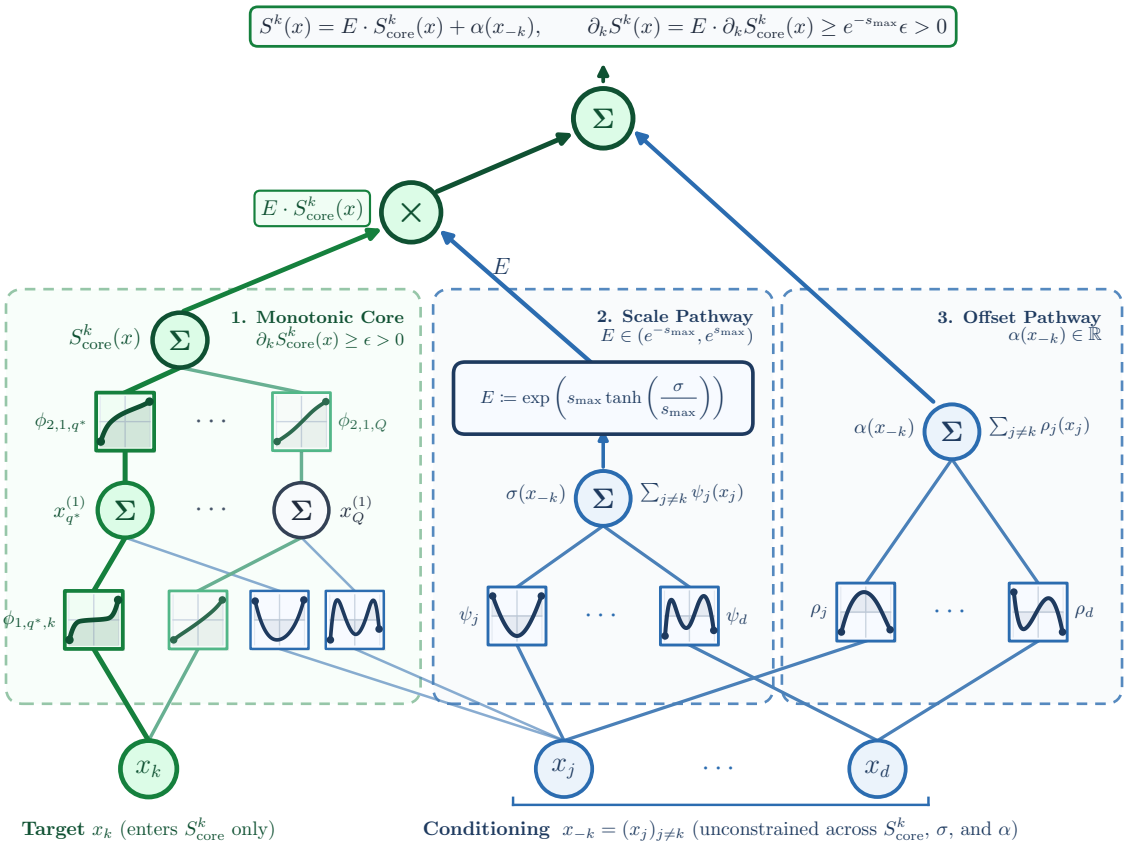


Figure 6: **Scale-Modulated Monotonic KAN (SM-KAN) Transport Map Architecture** (Definition 5.9). The target coordinate x_k (green) enters exclusively through the certified monotonic core network $S^k_{\text{core}}(x)$ (Module 1, left), where non-decreasing target-transmitting edges and an active path $\mathcal{P}^* = (k, q^*, 1)$ guarantee $\partial_k S^k_{\text{core}}(x) \geq \epsilon > 0$. Conditioning coordinates $x_{-k} = (x_j)_{j\neq k}$ (blue) modulate the map through three unconstrained pathways: internal conditioning edges within $S^k_{\text{core}}(x)$, an additive scale network $\sigma(x_{-k}) = \sum_{j\neq k} \psi_j(x_j)$ mapped through smooth saturation and exponentiation $E(x_{-k}) \coloneqq \exp(s_{\max} \tanh(\sigma(x_{-k})/s_{\max})) \in (e^{-s_{\max}}, e^{s_{\max}})$ to a multiplication gate ($\times$) computing $E(x_{-k}) \cdot S^k_{\text{core}}(x)$ (Module 2, center), and an additive offset network $\alpha(x_{-k}) = \sum_{j\neq k} \rho_j(x_j)$ summed at the output node ($\sum$) (Module 3, right).

Proposition 5.10 (Admissibility and Monotonicity Invariance Under Scale Modulation)**.** *Let S^k be parameterized according to Definition 5.9. Then,*

1. ***Target Monotonicity (Criterion 1):*** *For all $x \in \mathbb{R}^d$, the target derivative satisfies,*

$$\partial_k S^k(x) = E(x_{-k}) \cdot \partial_k S^k_{core}(x) \geq e^{-s_{\max}}\epsilon =: \epsilon_{SM} > 0, \tag{55}$$

where $\epsilon_{SM} \coloneqq e^{-s_{\max}}\epsilon > 0$ is a uniform positive constant, certifying that $S^k(x_{-k}, \cdot)$ is a 1D orientation-preserving diffeomorphism on $\mathbb{R}$.

2. ***Conditioning Freedom (Criterion 4):*** *For every conditioning coordinate $j \neq k$, the conditioning partial derivatives,*

$$\partial_j S^k(x) = E(x_{-k}) \left[\operatorname{sech}^2\left(\frac{\sigma(x_{-k})}{s_{\max}}\right) \psi_j'(x_j) \cdot S^k_{core}(x) + \partial_j S^k_{core}(x)\right] + \rho_j'(x_j), \tag{56}$$

$$\partial_j \partial_k S^k(x) = E(x_{-k}) \left[\operatorname{sech}^2\left(\frac{\sigma(x_{-k})}{s_{\max}}\right) \psi_j'(x_j) \cdot \partial_k S^k_{core}(x) + \partial_j \partial_k S^k_{core}(x)\right], \tag{57}$$

are unconstrained in sign.

Proof. Because $\sigma(x_{-k})$ and $\alpha(x_{-k})$ depend only on conditioning coordinates x_{-k}, they are constant with respect to target coordinate x_k, so $\partial_k[E(x_{-k})] \equiv 0$ and $\partial_k \alpha(x_{-k}) \equiv 0$. Differentiating (52) with respect to x_k yields,

$$\partial_k S^k(x) = E(x_{-k}) \cdot \partial_k S^k_{\text{core}}(x). \tag{58}$$

Since $\tanh(u) > -1$ for all $u \in \mathbb{R}$, we have $E(x_{-k}) > e^{-s_{\max}}$. Combined with $\partial_k S^k_{\text{core}}(x) \geq \epsilon > 0$ from Condition 1 of Definition 5.9, this yields the uniform bound,

$$\partial_k S^k(x) \geq e^{-s_{\max}} \epsilon =: \epsilon_{\text{SM}} > 0, \tag{59}$$

across all $x \in \mathbb{R}^d$, establishing Criterion 1.

Next, differentiating $S^k(x)$ in (52) with respect to a conditioning coordinate x_j $(j \neq k)$ via the product rule and chain rule yields,

$$\partial_j S^k(x) = \partial_j[E(x_{-k})] \cdot S^k_{\text{core}}(x) + E(x_{-k}) \cdot \partial_j S^k_{\text{core}}(x) + \partial_j \alpha(x_{-k}). \tag{60}$$

Applying the chain rule to the composite scale factor $E(x_{-k}) = \exp(s_{\max} \tanh(\sigma(x_{-k})/s_{\max}))$,

$$\partial_j[E(x_{-k})] = E(x_{-k}) \cdot \operatorname{sech}^2\left(\frac{\sigma(x_{-k})}{s_{\max}}\right) \cdot \partial_j \sigma(x_{-k}). \tag{61}$$

Substituting $\partial_j \sigma(x_{-k}) = \psi'_j(x_j)$ and $\partial_j \alpha(x_{-k}) = \rho'_j(x_j)$ from (53) and (54) and factoring out $E(x_{-k})$ yields (56). Differentiating $\partial_k S^k(x) = E(x_{-k}) \cdot \partial_k S^k_{\text{core}}(x)$ with respect to x_j likewise yields,

$$\begin{aligned} \partial_j \partial_k S^k(x) &= \partial_j[E(x_{-k})] \cdot \partial_k S^k_{\text{core}}(x) + E(x_{-k}) \cdot \partial_j \partial_k S^k_{\text{core}}(x) \\ &= E(x_{-k}) \left[\operatorname{sech}^2\left(\frac{\sigma(x_{-k})}{s_{\max}}\right) \psi'_j(x_j) \cdot \partial_k S^k_{\text{core}}(x) + \partial_j \partial_k S^k_{\text{core}}(x) \right], \end{aligned} \tag{62}$$

establishing (57). Because the univariate edge functions ψ_j and ρ_j are unconstrained in sign ($\theta \in \Theta$ in Definition 5.2), their derivatives $\psi'_j(x_j)$ and $\rho'_j(x_j)$ can take arbitrary real values of either sign independently of the core network. Consequently, $\partial_j S^k(x)$ and $\partial_j \partial_k S^k(x)$ are unconstrained in sign while strictly preserving target monotonicity $\partial_k S^k(x) \geq \epsilon_{\text{SM}} > 0$, establishing Criterion 4. □

6 Practical Enforcement Mechanisms

Having established in Section 5 that the two-layer monotonic KAN (Definition 5.5), the deep monotonic KAN (Definition 5.7), and the Scale-Modulated Monotonic KAN (Definition 5.9) satisfy all four admissibility criteria whenever their univariate edges obey the discrete-to-continuum monotonicity property (Definition 5.2), we now detail how we instantiate an edge family $\mathcal{F} = \{\phi_\theta\}_{\theta \in \Theta}$ using univariate B-splines:

1. **Target Monotonicity and Conditioning Freedom (Criterion 1 and Criterion 4):** We first verify that degree-p B-splines satisfy Definition 5.2 whenever their control points satisfy $c_{m+1} - c_m \geq \delta$, and we enforce this inequality via an *unconstrained cumulative softplus reparameterization* (Section 6.1).

2. **Global Real-Line Domain (Criterion 2):** To guarantee that these derivative bounds hold across all of $\mathbb{R}^d$ whenever input samples or intermediate hidden activations fall outside the finite knot grid $[t_{\min}, t_{\max}]$, we attach C^1*-continuous linear tail extrapolations* to every univariate spline (Section 6.2).

3. **Higher Regularity and Analytic Derivative Computability (Criterion 3 and Key Requirement 3):** We parameterize the univariate edges using B-splines of degree $p \geq 3$, which guarantees well-defined third-order derivatives for the generalized precision integrand $T_{jk}(x)$ and enables exact, closed-form evaluation of all required derivatives via B-spline recurrence (Section 6.3).

6.1 B-Spline Monotonicity and Cumulative Softplus Reparameterization

To instantiate Definition 5.2, we first record the classical property that discrete control-point monotonicity transfers to continuous derivative positivity for B-splines.

Lemma 6.1 (Monotonicity of 1D B-Splines)**.** *Let $\phi(t) = \sum_{m=1}^{M} c_m B_{m,p}(t)$ be a univariate B-spline of degree $p \geq 1$ defined over a non-decreasing knot sequence $t_1 \leq t_2 \leq \cdots \leq t_{M+p+1}$ with $t_{m+p} > t_m$ for all $m \in \{2, \ldots, M\}$. Then its first derivative on the active knot interval $[t_{p+1}, t_{M+1}]$ is the degree-$(p-1)$ B-spline expansion,*

$$\phi'(t) = p \sum_{m=2}^{M} \frac{c_m - c_{m-1}}{t_{m+p} - t_m} B_{m,p-1}(t). \tag{63}$$

If $c_m - c_{m-1} \geq 0$ for all $m \in \{2, \ldots, M\}$, then $\phi'(t) \geq 0$ on $[t_{p+1}, t_{M+1}]$. Moreover, if $c_m - c_{m-1} \geq \delta > 0$ for all $m \in \{2, \ldots, M\}$, then

$$\phi'(t) \geq \min_{2 \leq m \leq M} \left(\frac{p\,\delta}{t_{m+p} - t_m} \right) =: \epsilon > 0 \quad \forall t \in [t_{p+1}, t_{M+1}]. \tag{64}$$

Proof. On $[t_{p+1}, t_{M+1}]$, the basis functions $B_{m,p-1}(t)$ are non-negative and form a partition of unity ($\sum_{m=2}^{M} B_{m,p-1}(t) \equiv 1$) with positive knot spans $t_{m+p} - t_m > 0$. Consequently, $\phi'(t)$ in (63) is a convex combination of the scaled increments $p(c_m - c_{m-1})/(t_{m+p} - t_m)$, from which both lower bounds follow immediately. □

To enforce Lemma 6.1 without constrained optimization, we reparameterize the control vector $c = (c_1, \ldots, c_M)^\top \in \mathbb{R}^M$ of each monotonic B-spline in terms of an unconstrained base coefficient $c_1 \in \mathbb{R}$ and unconstrained increment parameters $\theta = (\theta_1, \ldots, \theta_{M-1})^\top \in \mathbb{R}^{M-1}$,

$$c_{m+1} = c_m + \mathrm{softplus}(\theta_m) + \delta_{\mathrm{mono}}, \quad m = 1, \ldots, M-1, \tag{65}$$

or equivalently in closed cumulative-sum form,

$$c_m = c_1 + \sum_{r=1}^{m-1} \Big(\mathrm{softplus}(\theta_r) + \delta_{\mathrm{mono}} \Big), \quad m = 2, \ldots, M, \tag{66}$$

where $\mathrm{softplus}(u) \coloneqq \log(1 + e^u) > 0$ and $\delta_{\mathrm{mono}} \geq 0$ is a fixed increment floor.

Because $\mathrm{softplus}(u) > 0$ for all $u \in \mathbb{R}$ with $\inf_{u \in \mathbb{R}} \mathrm{softplus}(u) = 0$, every control-point increment satisfies,

$$c_{m+1} - c_m = \mathrm{softplus}(\theta_m) + \delta_{\mathrm{mono}} > \delta_{\mathrm{mono}} \quad \forall \theta_m \in \mathbb{R}. \tag{67}$$

Along the active backbone path $\mathcal{P}^* \in \mathbb{P}_k$ of Definitions 5.5 and 5.7, setting a strictly positive floor $\delta_{\mathrm{mono}} = \delta > 0$ (e.g., $\delta_{\mathrm{mono}} = 10^{-4}$) places the spline in $\Theta_{\geq \epsilon}$ identically for all $(c_1, \theta) \in \mathbb{R}^M$, certifying $\phi'_{l,i_l^*,i_{l-1}^*}(t) \geq \epsilon_l > 0$ via Lemma 6.1. On the remaining target-transmitting splines where only non-decreasing slopes $\phi'_{l,j,i}(t) \geq 0$ are required ($\Theta_{\geq 0}$), one may set either $\delta_{\mathrm{mono}} = 0$ or $\delta_{\mathrm{mono}} > 0$. In contrast, for all conditioning inner splines $\phi_{1,j,i}$ ($i \neq k$) in Definition 5.7 and the scale and offset splines ψ_j, ρ_j ($j \neq k$) in Definition 5.9, no reparameterization is applied: their control vectors $c \in \mathbb{R}^M$ are updated directly as free parameters, preserving unrestricted sign and curvature for conditioning (Criterion 4). Because the map $(c_1, \theta) \mapsto c$ in (66) is smooth (C^∞) and bijective onto $\{c \in \mathbb{R}^M : c_{m+1} - c_m > \delta_{\mathrm{mono}}\}$, standard unconstrained optimizers such as Adam or L-BFGS can be applied directly without projection steps.

6.2 Global C^1 Linear Tail Extrapolation

To satisfy Criterion 2, the monotonicity guarantees of Section 5 must hold on all of $\mathbb{R}^d$, even when a spline input $t \in \mathbb{R}$, whether a raw sample coordinate x_i at layer $l = 1$ or an intermediate hidden activation $x_i^{(l-1)} = \sum_{r=1}^{n_{l-2}} \phi_{l-1,i,r}(x_r^{(l-2)})$ at layer $l \geq 2$, falls outside the active knot interval $[t_{\min}, t_{\max}] \coloneqq [t_{p+1}, t_{M+1}]$.

To guarantee well-posedness across all of $\mathbb{R}$, we extend every univariate spline ϕ outside $[t_{\min}, t_{\max}]$ via C^1-continuous linear tail extrapolation,

$$\phi_{\text{extrap}}(t) = \begin{cases} \phi(t_{\min}) + \phi'(t_{\min})(t - t_{\min}), & t < t_{\min}, \\ \phi(t), & t_{\min} \leq t \leq t_{\max}, \\ \phi(t_{\max}) + \phi'(t_{\max})(t - t_{\max}), & t > t_{\max}, \end{cases} \tag{68}$$

whose first derivative is globally continuous and constant on both exterior tails,

$$\phi'_{\text{extrap}}(t) = \begin{cases} \phi'(t_{\min}), & t < t_{\min}, \\ \phi'(t), & t_{\min} \leq t \leq t_{\max}, \\ \phi'(t_{\max}), & t > t_{\max}. \end{cases} \tag{69}$$

By Lemma 6.1 and the softplus reparameterization (65), the boundary slopes satisfy $\phi'(t_{\min}) \geq \epsilon_l > 0$ and $\phi'(t_{\max}) \geq \epsilon_l > 0$ for every spline along the active path $\mathcal{P}^*$, and $\phi'(t_{\min}) \geq 0$, $\phi'(t_{\max}) \geq 0$ for all remaining target-transmitting splines. Consequently, the derivative never vanishes in the tails, certifying that the B-spline family satisfies Definition 5.2 globally across all $t \in \mathbb{R}$.

6.3 Higher-Order Regularity and Closed-Form Derivative Recurrence

To satisfy Criterion 3, the transport map $S^k(x)$ must possess well-defined derivatives up to third order so that the generalized precision integrand $T_{jk}(x)$ in (21), which depends on the mixed third derivative $\partial_j \partial_k^2 S^k(x)$, can be evaluated without singular distributions. A univariate B-spline $\phi(t) = \sum_{m=1}^{M} c_m B_{m,p}(t)$ of polynomial degree p defined over simple interior knots is C^∞ on every open knot span (t_m, t_{m+1}) and belongs to continuity class C^{p-1} across the interior knots. Enforcing Criterion 3 therefore reduces to selecting a spline degree $p \geq 3$: cubic B-splines ($p = 3$) are C^2-continuous across interior knots with bounded, piecewise-constant third derivatives $\phi'''(t) \in L^\infty(\mathbb{R})$ (avoiding Dirac delta singularities at the knots, which have Lebesgue measure zero under any continuous density π), while degree $p \geq 4$ yields globally continuous third derivatives $\phi \in C^3([t_{\min}, t_{\max}])$ across the active knot domain.

Crucially, this B-spline parameterization simultaneously fulfills Key Requirement 3 (efficient derivative computability). Iterating the first-derivative recurrence (63) of Lemma 6.1 shows that the r-th derivative $\phi^{(r)}(t)$ ($1 \leq r \leq p$) of a degree-p B-spline on $[t_{\min}, t_{\max}]$ is itself an explicit B-spline of degree $p - r$,

$$\phi^{(r)}(t) = \sum_{m=r+1}^{M} c_m^{(r)} B_{m,p-r}(t), \quad \text{where} \quad c_m^{(r)} = \frac{p - r + 1}{t_{m+p-r+1} - t_m} \left(c_m^{(r-1)} - c_{m-1}^{(r-1)} \right), \tag{70}$$

initialized at $c_m^{(0)} = c_m$ (with $\phi_{\text{extrap}}^{(r)}(t) \equiv 0$ for all $r \geq 2$ on the linear tails $t \notin [t_{\min}, t_{\max}]$). Because each basis function $B_{m,p-r}(t)$ has compact support over $p-r+1$ knot intervals, evaluating $\phi(t)$ together with $\phi'(t)$, $\phi''(t)$, and $\phi'''(t)$ requires only $\mathcal{O}(p)$ arithmetic operations in closed form. In contrast to UMNNs, which must differentiate through a numerical quadrature rule via nested automatic differentiation, our B-spline KAN parameterization yields exact analytic expressions for all partial derivatives of $S^k(x)$, which we now exploit to compute the generalized precision matrix in closed form.

7 Exact Analytic Derivatives for the Generalized Precision Matrix

Evaluating the empirical generalized precision matrix $\widehat{\Omega}_{jk} = \frac{1}{N}\sum_{i=1}^{N} |T_{jk}(x^{(i)})|$ from (19) requires the mixed partial derivatives of the conditional log-likelihood up to third order. We adopt the following shorthand notation:

$$S_k \coloneqq \partial_k S^k(x), \qquad S_j \coloneqq \partial_j S^k(x), \tag{71}$$

$$S_{kk} \coloneqq \partial_k^2 S^k(x), \qquad S_{jk} \coloneqq \partial_j \partial_k S^k(x), \tag{72}$$

$$S \coloneqq S^k(x), \qquad S_{jkk} \coloneqq \partial_j \partial_k^2 S^k(x), \tag{73}$$

for all $j \neq k$. Differentiating $T_{jk}(x) = \partial_j \partial_k[-\frac{1}{2}S^2 + \log S_k]$ via the product and quotient rules yields the decomposition,

$$T_{jk}(x) = \underbrace{-S_j S_k - S S_{jk}}_{\text{Score Coupling Term}} + \underbrace{\frac{S_{jkk}S_k - S_{kk}S_{jk}}{S_k^2}}_{\text{Hessian Curvature Term}}. \tag{74}$$

7.1 Closed-Form Core KAN Derivatives

For the two-layer monotonic KAN core $S^k_{\text{core}}(x) = \sum_{q=1}^{Q} \phi_{2,1,q}(x_q^{(1)}(x))$ with $x_q^{(1)}(x) = \sum_{i=1}^{d} \phi_{1,q,i}(x_i)$ (Definition 5.5), repeated application of the chain and product rules expresses all five partial derivatives required by (74) directly in terms of the univariate B-spline derivatives $\phi^{(r)}$ ($r \in \{1,2,3\}$) from (70),

$$S_{k,\text{core}}(x) = \sum_{q=1}^{Q} \phi'_{2,1,q}\big(x_q^{(1)}\big)\, \phi'_{1,q,k}(x_k), \tag{75}$$

$$S_{j,\text{core}}(x) = \sum_{q=1}^{Q} \phi'_{2,1,q}\big(x_q^{(1)}\big)\, \phi'_{1,q,j}(x_j), \tag{76}$$

$$S_{jk,\text{core}}(x) = \sum_{q=1}^{Q} \phi''_{2,1,q}\big(x_q^{(1)}\big)\, \phi'_{1,q,j}(x_j)\, \phi'_{1,q,k}(x_k), \tag{77}$$

$$S_{kk,\text{core}}(x) = \sum_{q=1}^{Q} \Big[\phi''_{2,1,q}\big(x_q^{(1)}\big)\big(\phi'_{1,q,k}(x_k)\big)^2 + \phi'_{2,1,q}\big(x_q^{(1)}\big)\, \phi''_{1,q,k}(x_k)\Big], \tag{78}$$

$$S_{jkk,\text{core}}(x) = \sum_{q=1}^{Q} \phi'_{1,q,j}(x_j)\Big[\phi'''_{2,1,q}\big(x_q^{(1)}\big)\big(\phi'_{1,q,k}(x_k)\big)^2 + \phi''_{2,1,q}\big(x_q^{(1)}\big)\, \phi''_{1,q,k}(x_k)\Big]. \tag{79}$$

All five quantities are evaluated in closed form via vectorized tensor contractions across $j \in \{1,\dots,d\}\backslash\{k\}$ in a single forward pass, without numerical quadrature or automatic differentiation graphs.

7.2 Scale-Modulated Derivatives and the Hessian Cancellation Theorem

For the Scale-Modulated Monotonic KAN $S^k(x) = E \cdot S^k_{\text{core}}(x) + \alpha(x_{-k})$ (Definition 5.9), where $E \coloneqq E(\sigma(x_{-k})) = \exp(s_{\max}\tanh(\sigma(x_{-k})/s_{\max}))$, $\sigma_j \coloneqq \partial_j \log E = \operatorname{sech}^2(\sigma(x_{-k})/s_{\max})\psi'_j(x_j)$, and $\partial_j \alpha(x_{-k}) = \rho'_j(x_j)$ for $j \neq k$, differentiating with respect to x_k and x_j yields,

$$S_k \coloneqq E\, S_{k,\text{core}}, \qquad S_j \coloneqq E\big(\sigma_j S_{\text{core}} + S_{j,\text{core}}\big) + \rho'_j, \tag{80}$$

$$S_{kk} \coloneqq E\, S_{kk,\text{core}}, \qquad S_{jk} \coloneqq E\big(\sigma_j S_{k,\text{core}} + S_{jk,\text{core}}\big), \tag{81}$$

$$S \coloneqq E\, S_{\text{core}} + \alpha, \qquad S_{jkk} \coloneqq E\big(\sigma_j S_{kk,\text{core}} + S_{jkk,\text{core}}\big). \tag{82}$$

Theorem 7.1 (Hessian Cancellation Under Scale Modulation)**.** *Under the Scale-Modulated Monotonic KAN parameterization of Definition 5.9, the Hessian curvature term of the generalized precision integrand (74) is invariant to both the conditioning scale $\sigma(x_{-k})$ and the conditioning offset $\alpha(x_{-k})$,*

$$\frac{S_{jkk}S_k - S_{kk}S_{jk}}{S_k^2} = \frac{S_{jkk,core}S_{k,core} - S_{kk,core}S_{jk,core}}{(S_{k,core})^2}, \tag{83}$$

and the full precision integrand reduces to,

$$T_{jk}(x) = -S_jS_k - SS_{jk} + \frac{S_{jkk,core}S_{k,core} - S_{kk,core}S_{jk,core}}{(S_{k,core})^2}. \tag{84}$$

Proof. Substituting S_k, S_{kk}, S_{jk}, and S_{jkk} from (80)–(82) into the Hessian quotient gives,

$$\begin{aligned}\frac{S_{jkk}S_k - S_{kk}S_{jk}}{S_k^2} &= \frac{E\big(\sigma_j S_{kk,\text{core}} + S_{jkk,\text{core}}\big)\cdot ES_{k,\text{core}} - ES_{kk,\text{core}}\cdot E\big(\sigma_j S_{k,\text{core}} + S_{jk,\text{core}}\big)}{E^2(S_{k,\text{core}})^2} \\ &= \frac{E^2\Big(\sigma_j S_{kk,\text{core}}S_{k,\text{core}} + S_{jkk,\text{core}}S_{k,\text{core}} - \sigma_j S_{kk,\text{core}}S_{k,\text{core}} - S_{kk,\text{core}}S_{jk,\text{core}}\Big)}{E^2(S_{k,\text{core}})^2}.\end{aligned} \tag{85}$$

The scale-gradient cross-terms $\pm\sigma_j S_{kk,\text{core}}S_{k,\text{core}}$ cancel identically in the numerator, and canceling the common positive factor $E^2 > 0$ yields (83). Substituting (83) into (74) gives (84). □

7.3 Vectorized Precision Evaluation via Tensor Contractions

In our PyTorch implementation (`models/monotonic_kan/kan_map.py`), all core derivatives (75)–(79) and scale-modulated quantities (80)–(82) are evaluated simultaneously across a batch of B samples $x \in \mathbb{R}^{B\times d}$ using batched tensor contractions (`torch.einsum`):

1. **Spline Basis Evaluation:** For each variable coordinate $j \in \{1,\dots,d\}$, evaluate B-spline basis matrices:
$$b_{\text{val}},\ b_{d1},\ b_{d2},\ b_{d3} \in \mathbb{R}^{B\times d\times M}. \tag{86}$$

2. **Inner Layer Sums and Edge Derivatives:**
$$u_q = \texttt{torch.einsum("bdm,dqm->bq", b_val, inner_coeffs)}, \tag{87}$$
$$\phi'_{b,j,q} = \texttt{torch.einsum("bdm,dqm->bdq", b_d1, inner_coeffs)}, \tag{88}$$
$$\phi''_{b,j,q} = \texttt{torch.einsum("bdm,dqm->bdq", b_d2, inner_coeffs)}. \tag{89}$$

3. **Outer Spline Evaluations on Hidden Activations u_q:**
$$\Phi'_q = \texttt{torch.einsum("bqm,qm->bq", Phi_d1, outer_coeffs)}, \tag{90}$$
$$\Phi''_q = \texttt{torch.einsum("bqm,qm->bq", Phi_d2, outer_coeffs)}, \tag{91}$$
$$\Phi'''_q = \texttt{torch.einsum("bqm,qm->bq", Phi_d3, outer_coeffs)}. \tag{92}$$

4. **Vectorized Core Partial Derivatives:**
$$S_{k,\text{core}} = \texttt{torch.einsum("bq,bq->b", Phi_p, phi_k_p)}, \tag{93}$$
$$S_{j,\text{core}} = \texttt{torch.einsum("bq,bdq->bd", Phi_p, phi_prime)}, \tag{94}$$
$$S_{jk,\text{core}} = \texttt{torch.einsum("bq,bdq,bq->bd", Phi_pp, phi_prime, phi_k_p)}, \tag{95}$$
$$\begin{aligned}S_{kk,\text{core}} = {}&\texttt{torch.einsum("bq,bq->b", Phi_pp, phi_k_p**2)} \\ &+ \texttt{torch.einsum("bq,bq->b", Phi_p, phi_k_pp)},\end{aligned} \tag{96}$$
$$S_{jkk,\text{core}} = \texttt{torch.einsum("bdq,bq->bd", phi_prime, term_inner)}. \tag{97}$$

Because every derivative is computed in closed form across all conditioning dimensions $j \in \{1, \ldots, d\} \setminus \{k\}$ in a single vectorized pass, extracting the generalized precision matrix $\widehat{\Omega}$ completely bypasses autodiff graphs, achieving a 110.8× speedup over `torch.func.hessian` and an 84× to 300× speedup over UMNN numerical quadrature.

8 Summary, Empirical Benchmarks, and Conclusions

The experimental configurations and architectural specifications of the baseline UMNN and proposed Scale-Modulated Monotonic KAN (SM-KAN) are detailed in Table 1, while their computational efficiency, memory scaling, and structural expressivity are summarized in Table 2. All empirical benchmarks, exact numerical tables, and interactive demonstrations are implemented and verified in the companion Jupyter notebook `section8_benchmarks.ipynb`.

To provide a controlled evaluation, hardware profiling was conducted on an Intel Core i7-1065G7 CPU (1.30 GHz base, 3.90 GHz boost, 16 GB RAM) under PyTorch 2.14. The baseline UMNN parameterizes the strictly positive integrand $f(t, x_{-k}) > 0$ with a dense multi-layer perceptron comprising two hidden layers of $[16, 32]$ units, evaluated via Clenshaw–Curtis quadrature with $K = 50$ integration nodes per evaluation sample. Together with dense conditioning networks parameterizing the external scale and offset, this architecture contains 1,523 parameters per feature at dimension $d = 10$. In contrast, the Scale-Modulated KAN (SM-KAN) parameterizes $S^k(x) = E(x_{-k})S^k_{\text{core}}(x) + \alpha(x_{-k})$ by coupling a 2-layer monotonic KAN core ($[d \to Q \to 1]$ with hidden layer width $Q = 4$, 929 parameters) with two unconstrained conditioning networks for scale $E(x_{-k})$ and offset $\alpha(x_{-k})$ (100 parameters each), totaling 1,129 parameters per feature (1.35× more compact than UMNN, or 929 parameters and 1.64× more compact for the unmodulated additive core). All KAN edges employ cubic B-splines ($p = 3$) over an active knot domain $[t_{\min}, t_{\max}] = [-4.0, 4.0]$ partitioned into $G = 8$ intervals (11 control points per spline) with C^1 linear tail extrapolation, and the target core utilizes cumulative softplus control point parameterization ($\epsilon_{\text{mono}} = 10^{-4}$), eliminating numerical quadrature entirely.

Table 1: Architectural specifications and experimental configurations for baseline UMNN and Scale-Modulated KAN (SM-KAN). Benchmarked on an Intel Core i7-1065G7 CPU (1.30–3.90 GHz, 16 GB RAM) under PyTorch 2.14.

Specification	**UMNN (Baseline)**	**Scale-Modulated KAN (Ours)**
Architecture	MLP ($[16, 32]$) + Scale/Offset	2-Layer KAN ($Q = 4$) + Scale/Offset
Edge Basis	Smooth activations ($f > 0$)	Cubic B-splines ($p = 3$)
Knot Grid	None	Uniform $[-4.0, 4.0]$ ($G = 8$)
Tails	Standard feedforward tails	C^1 linear tail extrapolation
Monotonicity	Quadrature of $f > 0$	Cumulative softplus control points
Parameters ($d = 10$)	1,523	**1,129** (929 core +200 scale/offset)
Implementation	`section8_benchmarks.ipynb` (exact reproduction)	

Table 2: Head-to-head comparison between baseline UMNN and Scale-Modulated KAN (SM-KAN).

Metric / Property	**UMNN (Baseline)**	**Scale-Modulated KAN (Ours)**
Numerical Quadrature	Required (50 nodes/sample)	**None** (Exact closed-form)
Precision Evaluation ($d = 10$)	10.56 s	**0.14 s** (84× faster)
Precision Evaluation ($d = 40$)	203.6 s	**0.68 s** (299× faster)
Precision Evaluation ($d = 156$, Ovarian)	$\approx$ 42 min (projected)	**1.99 s** ($>$ 1200× faster)
Peak Memory Footprint ($d = 10$)	71.1 MB	**0.25 MB** (284× reduction)
Target Monotonicity ($\partial_k S^k > 0$)	Certified ($f > 0$)	**Certified** (Proposition 5.10)
Parameters per Feature ($d = 10$)	1,523	**1,129** (1.35× more compact)
Training Throughput	33,011 samples/s	**104,591 samples/s** (3.17× faster)
Multiplicative Dependencies	Supported (slow autodiff)	**Fully Supported** ($F_1 = 1.0$)

8.1 Benchmark Performance on Standard L-SING Testbeds

We evaluate SM-KAN against the baseline UMNN architecture across the three primary benchmark domains introduced in the original L-SING paper [7]:

1. **Gaussian Graphical Model ($d = 10$):** Evaluated on $M = 5{,}000$ training and $N = 2{,}000$ evaluation samples from a 10-dimensional Gaussian distribution with 4 ground-truth edges. At conservative threshold $\tau = 0.20$, both UMNN and SM-KAN achieve perfect recovery ($F_1 = 1.000, \text{FPR} = 0.000$). However, at threshold $\tau = 0.10$, UMNN degrades to $F_1 = 0.667$ with 4 false positives (FPR = 0.098), whereas SM-KAN maintains **perfect graph recovery ($F_1 = 1.000, \mathbf{FPR} = 0.000$)**. Training SM-KAN requires 28.1 s compared to 89.9 s for UMNN (3.2× faster), and precision matrix extraction completes in 0.14 s versus 10.56 s (84× faster).

2. **Non-Gaussian Butterfly Distribution ($d = 10$ and $d = 40$):** To evaluate heteroscedastic scale dependencies ($X_i \sim \mathcal{N}(0, 1)$, $Y_i = W_i X_i$), we test recovery on $d = 10$ (5 independent pairs) across sample sizes $M \in \{1{,}000, 5{,}000\}$. At $M = 5{,}000$, SM-KAN achieves $F_1 = 1.000$ and FPR = 0.000 across all thresholds $\tau \in [0.10, 0.30]$, maintaining a wide separation margin ($\Delta = 0.270$) with all spurious non-edges strictly suppressed below the $\tau = 0.10$ threshold (strongest non-edge at 0.091 versus weakest true edge at 0.362, yielding a 3.96× signal-to-noise contrast). At small sample size $M = 1{,}000$, UMNN incurs 18 false positives ($F_1 = 0.357, \text{FPR} = 0.450$), whereas SM-KAN maintains substantially higher precision ($F_1 = 0.625, \text{FPR} = 0.150$). Scaling to $d = 40$ (20 pairs, 780 undirected candidate edges), extracting the complete precision matrix takes **0.68 seconds** for SM-KAN compared to **203.6 seconds** for UMNN (299× speedup).

3. **TCGA Curated Ovarian Cancer Gene Network ($d = 156$):** Demonstrating applicability to high-dimensional biological networks, we train SM-KAN across 156 gene expression profiles from patient tumor samples ($N_{\text{trn}} = 346, N_{\text{val}} = 115, N_{\text{tst}} = 117$). All 156 conditional transport maps train in 254.9 s total, and evaluating the entire 156×156 generalized precision matrix across test patients takes only **1.99 seconds** with a memory footprint of 75.4 MB.

8.2 Summary of Contributions

In summary, the Scale-Modulated Monotonic KAN parameterization resolves the fundamental computational and architectural limitations of existing transport maps in L-SING:

1. **Elimination of Numerical Quadrature:** By replacing ODE and quadrature-based neural networks with edge basis families satisfying the discrete-to-continuum monotonicity property (Definition 5.2) and instantiating them with B-splines, SM-KAN computes forward

evaluations, functional losses, and generalized precision matrices in closed form, delivering 84× to 299× faster precision extraction.

2. **Unconditional Monotonicity by Construction:** Global target monotonicity $\partial_k S^k(x) \geq \epsilon > 0$ on $\mathbb{R}^d$ is certified for two-layer, deep L-layer, and scale-modulated KANs through cumulative softplus reparameterization of discrete B-spline control points and C^1 linear tail extrapolation, eliminating post-step projections.

3. **Resolution of the Algebraic Polarization Obstruction and Hessian Cancellation:** Multiplicative scale modulation $E(\sigma(x_{-k})) \cdot S^k_{\text{core}}(x) + \alpha(x_{-k})$ enables the map to represent multiplicative and heteroscedastic dependencies while preserving target monotonicity, and Theorem 7.1 proves that scale and offset modulation introduce no additional third-order curvature terms into the log-Jacobian component of the precision matrix.

4. **Higher-Order Regularity and Closed-Form Precision Derivatives:** Instantiating edges with B-splines of degree $p \geq 3$ satisfies the higher regularity required by the generalized precision integrand without numerical quadrature, admitting exact closed-form evaluation of all five partial derivatives and instantaneous extraction of the precision matrix.

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